SEASHORE FIBERGLASS LLC MASTER TECHNICAL MANUAL

Volume 1 — Fiberglass Deck Engineering & Construction

Chapter 1

Overview of Fiberglass Waterproof Deck Systems

Fiberglass waterproof deck systems are designed to create a seamless, durable, and long-lasting waterproof surface capable of protecting the structural framing and interior spaces below the deck.

Unlike traditional wood or composite decks, fiberglass decks function as both a walking surface and a waterproof membrane. This makes proper installation and detailing critical to long-term performance.

A complete fiberglass deck system typically consists of several integrated components:

- structural deck framing
- plywood substrate layers
- fiberglass reinforcement layer
- resin bonding system
- protective gelcoat finish

When installed correctly, this system creates a continuous waterproof shell that prevents water from penetrating the structure below.

Coastal Construction Considerations

Fiberglass decks are particularly common in coastal regions such as the Jersey Shore, where homes are exposed to environmental conditions that accelerate deterioration of traditional deck materials.

Key environmental factors include:

Salt Air

Salt carried by coastal winds can accelerate corrosion of metal components and degrade exposed materials.

High Humidity

Coastal environments experience consistently elevated humidity levels, which increases the risk of moisture retention in building materials.

UV Exposure

Continuous sunlight gradually degrades many building materials. Fiberglass systems rely on a protective gelcoat layer to shield the fiberglass reinforcement from ultraviolet damage.

Freeze-Thaw Cycles

Water that enters small cracks or joints may freeze during winter months, expanding and causing further damage.

Importance of Proper Waterproof Detailing

Fiberglass deck systems rely heavily on correct installation of waterproof transitions.

While the fiberglass surface itself provides the main waterproof membrane, many failures occur at transition areas where the deck meets other building components.

Critical transition areas include:

- wall connections
- door thresholds
- railing posts
- structural posts inside columns
- deck edges and drip edge transitions

Improper flashing at these locations can allow water to infiltrate beneath the fiberglass surface.

Surface Damage and Crack-Related Leaks

Although transition failures are a common source of leaks, water infiltration can also occur when the fiberglass surface itself becomes damaged.

Large cracks or deteriorated fiberglass areas may allow water to penetrate directly through the waterproof membrane.

If these cracks are not repaired in time, water may reach the plywood substrate beneath the fiberglass layer. Prolonged exposure to moisture can cause plywood deterioration, eventually leading to soft spots in the walking surface.

For this reason, surface cracks should always be inspected and repaired promptly to prevent structural damage to the deck system.

Chapter 2

Structural Deck Design and Subfloor Systems

The structural foundation of a fiberglass waterproof deck plays a critical role in the long-term performance of the system. Because fiberglass membranes are rigid waterproofing layers, the substrate beneath the fiberglass must remain stable and properly supported.

Any movement, flexing, or structural instability in the deck framing or plywood substrate can eventually lead to cracking in the fiberglass surface or deterioration of the waterproof system.

For this reason, proper deck framing, correct pitch, and a well-installed plywood substrate are essential components of fiberglass deck construction.

Deck Framing Conditions

Fiberglass decks are typically installed over conventional wood deck framing constructed with pressure-treated lumber.

The most common joist sizes encountered in deck construction include:

- **2x10 joists**
- **2x12 joists**

The exact size and spacing of joists are determined by the structural design of the deck and the architectural plans for the home.

Regardless of joist size, the framing must provide a stable and properly supported base before installation of the fiberglass system begins.

Importance of Deck Pitch

One of the most critical design elements of a waterproof deck is the pitch or slope of the deck surface.

Fiberglass decks must always be installed with sufficient slope to ensure that water drains away from the building structure.

The standard pitch used for fiberglass waterproof decks is approximately:

1/4 inch of slope per foot

This slope directs water away from walls and doors and toward the drainage edge or drainage system.

If the deck pitch is insufficient or improperly installed, water may accumulate on the deck surface. Standing water can accelerate wear of the fiberglass membrane and increase the risk of leaks over time.

Correcting Improper Deck Pitch

In many existing decks, the original framing may not provide adequate pitch for proper drainage. In these situations, the pitch must be corrected before installing or restoring the fiberglass system.

Two common methods are used to correct deck pitch:

- sleepers installed above existing joists
- sister joists installed alongside existing joists

While both methods may be used depending on the situation, sister joists are often preferred in many repair scenarios.

This is because existing deck joists may become bowed or sagged over time. If sleepers are installed directly on top of bowed joists, the resulting deck surface may continue to follow the same deformation.

By installing new **sister joists**, contractors can create a new straight structural reference that corrects the pitch and eliminates the existing sag in the framing.

Plywood Subfloor System

The plywood substrate serves as the direct base for the fiberglass waterproof membrane.

In fiberglass deck construction, the plywood layers must provide a rigid, stable surface that minimizes movement under load.

A common and reliable subfloor configuration consists of two plywood layers:

First layer:
- 3/4 inch CDX plywood
- glued to the framing
- fastened with **3 1/4 inch galvanized nails**

This layer provides the primary structural deck surface.

Second layer:
- 1/2 inch ACX plywood
- glued and screwed to the first layer

This second layer creates a smooth and stable substrate for fiberglass installation.

Using screws for the second layer significantly reduces movement in the plywood system and helps prevent nail pops or surface irregularities.

Importance of Glue and Fastening

The use of construction adhesive between plywood layers greatly increases the rigidity of the deck surface.

By bonding the two layers together, the plywood system behaves more like a single structural panel rather than two independent sheets.

Fastening patterns using nails for the first layer and screws for the second layer help ensure that the deck surface remains stable under load and over time.

Reducing movement in the plywood substrate is essential for preventing stress cracks in the fiberglass membrane.

Structural Stability and Long-Term Performance

Fiberglass waterproof deck systems depend on the structural stability of the underlying deck.

If the framing or plywood layers move excessively, the fiberglass membrane may eventually develop cracks or delamination.

Proper framing, correct pitch, and secure fastening of plywood layers help ensure that the fiberglass system performs as intended for many years.

For this reason, structural evaluation of the deck framing and substrate is always an important step before installing or repairing fiberglass waterproof decks.

Chapter 3

Perimeter Edge Engineering for Open Deck Systems

The perimeter edge of a fiberglass waterproof deck is one of the most critical areas of the entire system. Proper edge construction ensures that water is directed away from the structure and prevents moisture from reaching structural components of the deck.

In coastal environments, where decks are constantly exposed to rain, humidity, and salt air, improper edge detailing can quickly lead to water infiltration and structural damage.

For this reason, a well-designed perimeter system is essential for long-term waterproof performance.

Seashore Fiberglass uses a specific perimeter construction method designed to control water flow and integrate properly with exterior trim and siding systems.

Standard Perimeter Build-Up for Open Decks

Open decks typically drain water toward the perimeter edge. Because of this, the edge assembly must be carefully designed to ensure that water flows outward and away from the structure.

The standard perimeter build-up used for open fiberglass decks consists of the following sequence:

1. Installation of 5/4 pressure-treated packer boards
2. Installation of perimeter metal flashings
3. Installation of plywood substrate
4. Installation of PVC drip edge

This layered approach creates a controlled water path while also allowing proper integration with exterior siding and fascia components.

Packer Board Installation

The first component installed along the deck perimeter is the packer board.

Packer boards are typically constructed from **5/4 pressure-treated lumber**, with an actual width of approximately **1¾ inches**.

These packer boards are installed directly on top of the joists around the entire deck perimeter.

The packer board serves several important purposes:

• acts as a narrow spacer for the deck edge
• creates the correct offset for exterior fascia installation
• supports the plywood edge at the perimeter
• helps define the correct water drainage path

Because the packer board is narrower than the exterior drip edge and fascia components, it allows these exterior elements to overlap properly without forcing or misalignment.

Integration with Exterior Fascia Systems

In most residential construction projects, the exterior fascia boards are installed later by the siding or trim crew.

The packer board system is specifically designed to make this integration simple and effective.

Because the packer board is narrower than the PVC drip edge, it creates a small channel that allows the exterior fascia board to slide easily underneath the deck edge.

This prevents gaps between the deck edge and the exterior trim while maintaining proper alignment of the materials.

Proper integration between the deck edge and the fascia system helps prevent water from entering the structure through the perimeter joint.

Metal Flashing Installation

After the packer boards are installed, metal flashing is typically installed along the deck perimeter and drip edge ends.

These metal flashings provide an additional layer of protection and help integrate the deck waterproofing system with the exterior building envelope.

The flashing allows housewrap or siding paper installed later by the siding contractor to slide underneath the flashing layer.

This ensures that the waterproofing layers remain properly layered in the correct sequence.

By placing the fiberglass flashing beneath siding materials, water is always directed outward and away from the structure.

Plywood Installation Over Packer Boards

Once the perimeter flashing is installed, the plywood substrate is installed across the entire deck surface.

The plywood layer extends over the packer boards, fully covering them.

This step is important because it prevents the packer board from being exposed to moisture.

By covering the packer with plywood, the wood spacer becomes fully protected within the deck assembly.

This design allows the packer board to serve as a structural spacer without ever being exposed to direct water contact.

PVC Drip Edge Installation

After the plywood substrate is installed, the final perimeter component is the **PVC drip edge**.

Seashore Fiberglass typically uses **1x4 PVC drip edge**, which has an approximate width of **3½ inches**.

The PVC drip edge is significantly wider than the packer board.

Because of this difference in width, the drip edge extends beyond the packer and creates a proper overhang that directs water away from the structure.

The PVC material provides several advantages:

- completely rot-proof
- resistant to moisture exposure
- durable in coastal environments
- compatible with fiberglass waterproofing systems

PVC drip edge also provides a clean and durable edge for the fiberglass membrane to terminate.

Water Flow Control

The perimeter system is designed so that water always follows a controlled path.

When rainwater reaches the edge of the deck, it flows:

- across the fiberglass surface
- over the PVC drip edge
- over the exterior fascia installed below

Because the packer board remains protected behind the plywood and drip edge, it is never exposed to direct water contact.

This prevents moisture from entering the structure and helps eliminate the risk of rot in the deck edge components.

Advantages of the Packer Board Edge System

The packer board edge system provides several important benefits:

• improves integration with exterior trim and siding systems
• prevents exposed wood at the deck edge
• creates a controlled water drainage path
• simplifies installation for trim contractors
• increases long-term durability of the deck edge

By carefully controlling the sequence of materials at the deck perimeter, this system helps ensure reliable waterproof performance in demanding coastal conditions.

Chapter 4

Fiberglass Waterproof Membrane Installation

The fiberglass waterproof membrane forms the primary protective layer of a fiberglass deck system. This membrane creates a seamless waterproof barrier that prevents water from penetrating the plywood substrate and structural framing beneath the deck.

Proper installation of the fiberglass membrane is critical to ensure long-term durability and waterproof performance.

The fiberglass system typically consists of:

• fiberglass reinforcement mat
• resin bonding system
• protective gelcoat finish

When installed correctly, these components form a continuous waterproof shell across the entire deck surface.

Surface Preparation

Before fiberglass can be installed, the plywood substrate must be properly prepared to ensure strong adhesion.

Surface preparation begins by grinding the entire deck surface. Grinding removes surface irregularities, smooths joints between plywood panels, and improves mechanical bonding between the fiberglass resin and the plywood substrate.

Grinding also helps remove minor surface contaminants that could interfere with proper bonding.

After grinding is complete, the deck surface should be thoroughly cleaned to remove dust and debris.

Proper surface preparation is essential for achieving a strong and reliable fiberglass bond.

Sealing Screw Heads and Plywood Joints

After surface preparation, screw heads, fastener locations, and plywood seams must be sealed.

This is typically done using a body filler material such as Bondo.

The purpose of filling these areas is to eliminate surface imperfections that could telegraph through the fiberglass layer.

Sealing screw heads and joints also helps prevent resin absorption differences that could lead to uneven fiberglass curing.

Areas commonly filled include:

- screw heads
- plywood seams
- minor surface cracks
- localized imperfections

Once the filler material has cured, the filled areas are lightly sanded or ground to create a smooth surface.

Fiberglass Reinforcement Layer

The fiberglass reinforcement layer is installed across the entire deck surface.

Seashore Fiberglass typically uses **2-ounce fiberglass mat** as the reinforcement material.

Fiberglass mat consists of randomly oriented glass fibers that provide strength and flexibility when saturated with resin.

The mat is carefully positioned across the deck surface and fully saturated with resin during installation.

The resin penetrates the fiberglass fibers and bonds the reinforcement layer to the plywood substrate.

Once cured, this composite layer forms a strong and continuous waterproof membrane.

Resin Saturation

Resin application is a critical step in the fiberglass installation process.

The goal of resin saturation is to ensure that all fiberglass fibers are completely embedded within the resin matrix.

Proper saturation ensures that:

- no dry fiberglass fibers remain exposed
- the fiberglass layer bonds fully to the plywood substrate
- the membrane forms a uniform and continuous surface

Technicians must carefully roll the resin into the fiberglass mat to remove air pockets and ensure complete saturation.

Air bubbles or dry spots can weaken the membrane and reduce long-term durability.

Curing Process

Once the fiberglass mat has been fully saturated with resin, the material must be allowed to cure.

During curing, the resin hardens and bonds the fiberglass reinforcement layer to the deck surface.

The curing time depends on several factors including:

- temperature
- humidity
- resin formulation

Proper curing is necessary before any additional surface preparation or finishing steps can be performed.

Grinding the Fiberglass Surface

After the fiberglass layer has fully cured, the surface must be ground again.

This step serves several purposes:

• smooths irregularities in the fiberglass surface
• removes high spots or rough texture
• prepares the surface for gelcoat adhesion

Grinding the fiberglass surface is a critical preparation step before applying the gelcoat finish.

Without proper surface preparation, the gelcoat layer may not adhere correctly.

Surface Cleaning with Acetone

After grinding the fiberglass surface, the entire deck must be cleaned thoroughly.

Acetone is commonly used to remove dust, oils, and other contaminants from the fiberglass surface.

This cleaning step ensures that the gelcoat layer will bond properly to the fiberglass membrane.

Any residue left on the surface can compromise the adhesion of the gelcoat.

Gelcoat Application

The final step in the fiberglass waterproofing process is the application of gelcoat.

Gelcoat serves as the protective outer layer of the deck system.

Functions of gelcoat include:

• UV protection
• moisture resistance
• durability under foot traffic
• slip resistance when textured

Seashore Fiberglass typically applies gelcoat using **roller application**, which allows installers to control thickness and texture.

A textured gelcoat finish is commonly used to provide a non-slip walking surface.

The gelcoat layer seals the fiberglass membrane and provides the final finished appearance of the deck.

Continuous Waterproof Surface

Once the gelcoat has cured, the fiberglass deck system becomes a seamless waterproof surface.

Because fiberglass forms a continuous membrane, there are no seams where water can penetrate the system.

When properly installed and maintained, fiberglass waterproof decks can provide long-term protection for both exterior surfaces and interior living spaces below.

Chapter 5

Flashing Systems and Transition Waterproofing

In fiberglass waterproof deck systems, transitions between the deck surface and other building components represent some of the most vulnerable areas for water infiltration.

While the fiberglass field membrane creates a continuous waterproof surface across the deck, water can enter the system if transition details are not properly designed and installed.

For this reason, proper flashing installation is critical for maintaining the integrity of the waterproof system.

Key transition areas that require careful flashing integration include:

• walls connected to the deck
• door thresholds and door frames
• railing posts and structural mount posts

• structural posts located inside decorative columns
• roof or siding transitions connected to the deck surface

When these areas are not properly flashed, water may infiltrate beneath the fiberglass membrane and begin damaging the plywood substrate and structural framing.

Wall Flashing Systems

Where a fiberglass deck meets a vertical wall, a continuous flashing system must be installed to prevent water from entering the wall assembly.

In properly installed fiberglass deck systems, the fiberglass flashing should extend vertically up the wall surface.

A typical flashing height used in fiberglass deck installations is approximately:

12 inches above the deck surface

This vertical extension ensures that water flowing across the deck cannot reach the top edge of the flashing.

During new construction projects, fiberglass flashing is typically installed before siding materials are applied. This allows the flashing to remain beneath the siding layers, creating a properly layered waterproof barrier.

In cases where siding has already been installed, repair work may require temporarily removing sections of siding to access the wall area and install or repair the fiberglass flashing system.

Door Flashing Systems

Door thresholds represent another critical transition point in fiberglass deck systems.

If flashing around the door frame is missing or improperly installed, water can easily penetrate beneath the door and enter the building structure.

When correcting door flashing problems, contractors may need to temporarily remove the door and surrounding trim in order to access the framing and install proper waterproofing.

A typical door flashing repair may include:

• removing interior and exterior trim around the door
• temporarily removing the door unit

• repairing any rotted plywood or framing beneath the threshold
• installing fiberglass flashing around the door frame
• installing flashing tapes integrated with the housewrap system
• reinstalling the door and trim components

Proper integration between fiberglass flashing and the building envelope helps ensure that water is directed outward and away from the interior structure.

Flashing Around Railing Posts

Posts used for deck railings often penetrate the waterproof surface of fiberglass decks.

Common railing post types include:

• traditional 4x4 pressure-treated posts
• structural mount posts used with modern railing systems

These posts are frequently covered with decorative vinyl sleeves that match the railing system.

Regardless of post type, any penetration through the deck surface must be properly waterproofed.

If posts are installed without proper flashing, water may enter through the fastener penetrations and begin damaging the plywood substrate beneath the fiberglass membrane.

To prevent this issue, fiberglass flashing should be installed around the base of each post.

Fiberglass flashing typically extends approximately:

6 inches vertically around the post

This flashing creates a waterproof seal between the post and the deck surface.

In some cases where railing posts have been improperly installed, repair work may require removing the existing posts, installing structural blocking between joists, reinstalling plywood, and re-flashing the post base with fiberglass.

Structural Posts Inside Decorative Columns

Structural posts supporting decks or roof structures are sometimes concealed inside decorative column covers.

These column covers are often constructed using PVC or Azek materials and include trim components that hide the structural post inside.

Because the structural post is hidden within the column assembly, flashing problems in this area may remain undetected for long periods of time.

If fiberglass flashing is missing or improperly installed around the structural post, water can infiltrate the deck system and cause localized plywood deterioration.

This type of damage often appears as soft spots around the base of the column.

When investigating suspected leaks around columns, the lower trim components of the column may be removed in order to inspect the flashing condition around the structural post.

If necessary, repairs may include:

- removing damaged plywood
- installing structural blocking where needed
- reinstalling plywood
- installing proper fiberglass flashing around the structural post

Proper flashing installation ensures that water cannot enter the deck system at this critical structural penetration.

Other Wall Penetrations

In some cases, leaks near deck walls may originate from building components installed in the wall rather than from the fiberglass deck system itself.

Common wall penetrations include:

- electrical outlets
- exterior lighting fixtures
- water supply connections
- gas lines

If these penetrations are not properly flashed by the siding contractor, water may infiltrate behind the siding and travel downward into the deck area.

When these conditions are identified, corrective measures may include installing flashing tape and housewrap integration around the wall penetration.

Although these repairs involve the building envelope rather than the fiberglass deck system itself, correcting these issues helps prevent water from reaching the deck structure.

Importance of Proper Flashing Integration

Proper flashing integration ensures that all waterproofing layers are installed in the correct sequence.

The fundamental principle of waterproof construction is that each layer must overlap the layer below it in a way that directs water outward and away from the structure.

When fiberglass flashing, siding materials, housewrap, and drip edges are properly layered, water is safely directed away from the building.

Failure to maintain this layering sequence can allow water to travel behind the waterproof system and cause hidden structural damage.

Chapter 6

Deck Restoration Systems: Reglass and Recolor

Over time, fiberglass waterproof decks may experience surface wear, cosmetic deterioration, or localized damage caused by environmental exposure, structural movement, or previous installation errors.

Rather than replacing the entire deck system, many fiberglass decks can be restored through specialized restoration methods.

Two of the most common restoration systems used in fiberglass deck maintenance are:

- **Reglass systems**
- **Recolor systems**

Each method addresses a different level of deck deterioration.

Selecting the correct restoration method requires careful inspection of the existing fiberglass surface and underlying substrate.

Reglass Deck Systems

Reglassing is a restoration process in which a new fiberglass layer is installed over an existing deck system.

This method is typically used when the existing fiberglass surface shows significant deterioration but the structural deck framing remains stable.

Common situations where reglassing may be recommended include:

- widespread cracking in the fiberglass surface
- aging fiberglass that has lost structural integrity
- recurring leaks caused by deteriorated fiberglass
- deck surfaces that have undergone multiple previous repairs

Reglassing allows the deck to receive a completely new waterproof membrane without requiring full structural reconstruction.

Surface Stabilization Before Reglass

Before a new fiberglass layer can be installed, the existing deck substrate must be stabilized.

Older fiberglass decks were often constructed using nails rather than screws to fasten plywood to the framing. Over time, these nails may loosen, causing the plywood surface to move slightly under foot traffic.

This movement can eventually cause fiberglass cracks and surface damage.

To correct this condition, the deck surface is typically re-secured by installing screws through the existing fiberglass surface and into the joists below.

These screws are installed along each joist across the entire deck surface.

Although this process penetrates the existing fiberglass membrane, the new fiberglass layer that will be installed during reglassing will fully restore the waterproof barrier.

Re-securing the plywood substrate greatly improves deck stability and helps prevent future cracking in the fiberglass system.

Drip Edge Replacement

During reglass projects, existing drip edges are often inspected for deterioration.

Older decks sometimes used wooden drip edges that may have deteriorated due to long-term moisture exposure.

When necessary, deteriorated drip edges are removed and replaced with **PVC or Azek drip edge components**, which are rot-resistant and better suited for coastal environments.

Replacing damaged drip edges helps ensure proper water drainage at the perimeter of the deck.

Flashing Corrections During Reglass

Many fiberglass deck leaks originate from improperly installed flashing at transition areas.

For this reason, reglass projects often include inspection and correction of flashings around critical transition points such as:

- walls connected to the deck
- door thresholds
- railing posts
- structural posts located inside columns

If flashing problems are identified, siding materials or trim components may be temporarily removed to allow installation of proper fiberglass flashings.

Correcting flashing problems during reglass work helps eliminate the root cause of many leaks.

Grinding the Existing Fiberglass Surface

Once structural stabilization and flashing corrections are complete, the existing fiberglass surface must be prepared to receive the new fiberglass layer.

The entire deck surface is thoroughly ground to create a mechanical bonding surface.

Grinding removes surface contaminants and roughens the existing fiberglass, allowing the new fiberglass layer to adhere properly.

This step is critical to ensure long-term bonding between the old fiberglass surface and the new reinforcement layer.

Installation of New Fiberglass Layer

After grinding and preparation are complete, a new fiberglass reinforcement layer is installed across the entire deck surface.

As with new fiberglass installations, the reinforcement typically consists of **2-ounce fiberglass mat saturated with resin**.

This new fiberglass layer creates a completely renewed waterproof membrane across the deck.

Because the entire deck receives a new fiberglass layer, previous penetrations and minor surface defects are fully encapsulated within the new waterproof system.

Gelcoat Application

Once the new fiberglass layer has cured, the surface is ground again to prepare it for finishing.

The deck surface is then cleaned thoroughly using acetone to remove dust and contaminants.

Finally, a new **textured gelcoat finish** is applied to protect the fiberglass membrane and provide a durable, slip-resistant walking surface.

This gelcoat layer also restores the visual appearance of the deck.

Recolor Deck Systems

Recoloring is a lighter restoration method used when the fiberglass membrane remains structurally sound but the surface finish has deteriorated.

In these cases, the fiberglass layer itself remains intact, and only the protective gelcoat surface requires renewal.

Recolor systems typically involve:

- grinding the existing gelcoat surface
- repairing minor cracks or damaged areas with fiberglass patches
- smoothing and blending the repaired areas

• cleaning the surface with acetone
• applying a new gelcoat finish

Recoloring restores the protective coating of the deck without requiring installation of a new fiberglass reinforcement layer.

Identifying When Reglass Is Necessary

During recolor preparation, technicians may discover damage that indicates a deeper problem with the fiberglass membrane.

Examples include:

• large cracks in the fiberglass surface
• widespread structural deterioration of fiberglass
• water infiltration through the fiberglass membrane

In these cases, recoloring alone may not be sufficient.

When the fiberglass membrane itself is compromised, a **full reglass system** is typically recommended to restore the waterproof integrity of the deck.

Continuous Waterproof Restoration

Both reglass and recolor systems are designed to extend the life of fiberglass waterproof decks.

By properly evaluating the condition of the deck and selecting the appropriate restoration method, contractors can restore waterproof performance while avoiding unnecessary structural reconstruction.

Routine inspection and timely repairs are key to maintaining fiberglass deck systems and preventing long-term structural damage.

Chapter 7

Soft Spot Diagnosis and Repair Systems

Soft spots are one of the most common indicators of structural deterioration in fiberglass waterproof deck systems. A soft spot occurs when the plywood substrate beneath the

fiberglass membrane becomes weakened or deteriorated, causing the walking surface to feel unstable or compress under foot pressure.

Although the fiberglass membrane itself may appear intact in some cases, underlying moisture damage can compromise the structural integrity of the deck surface.

Early detection and proper repair of soft spots are critical to preventing further structural damage.

Common Causes of Soft Spots

Soft spots typically develop when water infiltrates beneath the fiberglass membrane and reaches the plywood substrate.

Over time, prolonged moisture exposure can cause the plywood to deteriorate, leading to structural weakening of the deck surface.

Common causes of water infiltration include:

- improperly installed fiberglass flashings
- missing flashing around walls or doors
- improperly sealed railing post penetrations
- missing or deteriorated flashing around structural posts inside columns
- large cracks in the fiberglass surface that allow water infiltration

In many cases, soft spots develop near transition areas where waterproofing details are more complex.

Soft Spots Around Columns

One of the most common locations for soft spot development is around decorative columns that conceal structural posts.

Structural posts are often wrapped with PVC or Azek column covers for aesthetic purposes. Because the structural post is hidden within the column assembly, flashing defects may remain undetected for long periods.

If fiberglass flashing around the structural post is missing or improperly installed, water can penetrate the deck system and gradually damage the plywood around the column base.

This damage may initially appear as a small soft area near the column but can expand over time if not addressed.

Field Detection of Soft Spots

Soft spots are often detected during routine inspection or when homeowners report unusual movement in the deck surface.

One of the most effective methods for detecting soft spots is simply walking across the deck surface and applying pressure with body weight.

Technicians look for areas where the deck surface feels:

- soft or spongy under foot
- unstable compared to surrounding areas
- visibly depressed or sagging

In advanced cases, the damaged area may be visible as a slight depression in the deck surface.

Inspection around columns, posts, and transition areas is particularly important because these locations are more susceptible to water infiltration.

Inspection of Column Flashings

When soft spots are detected near columns, it may be necessary to inspect the flashing condition around the structural post inside the column assembly.

Many decorative columns include removable trim components near the base.

By removing the lower trim pieces, technicians may gain access to the interior of the column and inspect the flashing condition around the structural post.

If flashing defects are found, corrective repairs can be performed to restore the waterproof integrity of the deck system.

Localized Soft Spot Repair Procedure

When soft spots are isolated to a limited area of the deck, localized repair is often possible without rebuilding the entire deck system.

Localized repair typically involves the following steps:

1. Cutting the fiberglass surface around the damaged area.
2. Removing the deteriorated plywood substrate.
3. Inspecting the framing beneath the plywood for additional damage.
4. Replacing the damaged plywood with new plywood of the same thickness as the original layers.
5. Securing the replacement plywood using appropriate fastening methods.
6. Grinding the repaired area to prepare for fiberglass bonding.
7. Filling screw heads and joints with body filler such as Bondo.
8. Installing new fiberglass reinforcement over the repaired section.
9. Grinding the fiberglass patch after curing.
10. Cleaning the area with acetone.
11. Applying gelcoat to restore the finished surface.

When decks contain multiple plywood layers, both layers may need to be replaced if moisture damage has affected the entire substrate thickness.

Structural Evaluation During Repair

During soft spot repairs, technicians must carefully evaluate the condition of the deck framing beneath the plywood.

If water infiltration has been present for an extended period, structural joists or supporting components may also have been affected.

In such cases, additional structural repairs may be required before reinstalling the plywood and fiberglass system.

Identifying and correcting the source of water infiltration is an essential part of any soft spot repair.

Without correcting the underlying cause of the leak, the problem may return even after the surface repair is completed.

Preventing Future Soft Spot Development

Preventing soft spots requires maintaining the waterproof integrity of the fiberglass deck system.

Routine inspection and timely repairs can significantly reduce the risk of moisture-related damage.

Key preventative measures include:

• repairing fiberglass cracks before water infiltration occurs
• maintaining proper flashing at walls, doors, and posts
• ensuring that railing post penetrations remain properly sealed
• monitoring deck edges and drip edges for deterioration

Regular maintenance and proper waterproof detailing help extend the service life of fiberglass waterproof decks.

Chapter 8

Structural Joist Repair Systems

In fiberglass waterproof deck systems, prolonged water infiltration can eventually affect not only the plywood substrate but also the structural framing beneath the deck.

When water penetrates the fiberglass membrane and remains trapped for extended periods, the wooden joists supporting the deck may begin to deteriorate.

Structural joist damage represents a more advanced stage of deck deterioration and requires careful repair procedures to restore the structural integrity of the deck.

Because joists carry the structural loads of the deck and any structures above it, repairs must be performed with proper support and safety measures.

Causes of Structural Joist Damage

Structural joists typically deteriorate when water infiltration occurs over long periods of time without being detected.

Common causes of water infiltration that may lead to joist damage include:

- improperly installed fiberglass flashings
- missing flashing around structural posts inside columns
- unsealed railing post penetrations
- large cracks in the fiberglass membrane
- prolonged soft spot conditions that allow water to reach framing

When moisture remains trapped within the deck assembly, the structural lumber may begin to rot and lose its load-bearing capacity.

Inspection and Evaluation

When soft spots or visible damage suggest possible structural deterioration, a thorough inspection of the framing is required.

In many deck constructions, the underside of the deck is covered by a soffit or ceiling system.

To access the framing, technicians may need to remove sections of the soffit below the deck.

Removing the soffit allows technicians to visually inspect the condition of the joists, beams, and supporting components.

During inspection, technicians evaluate:

- wood integrity
- signs of rot or fungal decay
- structural deformation
- compromised fasteners or connections

Temporary Structural Support

Before removing any damaged joists, temporary structural support must be installed.

Temporary support walls are commonly constructed using **2x4 framing** placed beneath the affected section of the deck.

These temporary supports carry the load of the deck and any structures above it while repairs are being performed.

Proper temporary support is essential to prevent structural collapse during the repair process.

Removal of Damaged Joists

Once temporary supports are in place, the deteriorated joists can be carefully removed.

Joists are typically constructed from pressure-treated lumber, commonly in sizes such as:

- 2x10
- 2x12

The damaged joist sections are cut out and removed without disturbing adjacent framing components whenever possible.

In cases where multiple joists have been affected, larger sections of framing may need to be replaced.

Installation of Replacement Joists

Replacement joists must match the size and structural capacity of the original framing members.

New **pressure-treated joists** are installed in the same configuration as the original joists, following local building codes and structural requirements.

The new joists are securely fastened to existing beams, ledger boards, and supporting structures.

Ensuring proper alignment and spacing of replacement joists is essential for restoring structural integrity and maintaining a stable deck surface.

Additional Structural Repairs

In some cases, structural damage may extend beyond the joists.

Additional components that may require repair include:

- support posts
- beams
- plywood substrate
- blocking between joists

If structural posts or other framing elements have been compromised by moisture damage, they must be repaired or replaced before rebuilding the deck surface.

Proper evaluation of all affected components helps ensure that the deck structure is fully restored.

Reinstallation of Deck Components

Once the structural repairs have been completed, the deck assembly can be rebuilt.

This process typically includes:

• reinstalling plywood substrate layers
• reinstalling fiberglass waterproofing systems
• restoring flashing systems at transition points

If the underside of the deck included a soffit or ceiling system, those components are reinstalled after structural repairs are complete.

Exterior fascia components may also be removed and reinstalled as part of the repair process.

Ensuring Long-Term Structural Performance

Structural joist repairs must restore the deck to a condition that is equal to or stronger than the original construction.

Careful attention must be given to:

• proper framing alignment
• secure fastening methods
• adequate drainage and waterproofing

By correcting both the structural damage and the original source of water infiltration, the repaired deck can maintain long-term performance and structural safety.

Chapter 9

Railing Post Systems and Waterproof Integration

Railing posts are one of the most common penetrations found in fiberglass waterproof deck systems. Because these posts often penetrate the waterproof membrane, they represent a potential entry point for water if not properly installed and sealed.

In coastal regions such as the Jersey Shore, railing systems are commonly installed using modern railing products that include decorative vinyl sleeves and composite rail assemblies. While these systems provide an attractive finish, the structural components beneath the decorative covers must be properly integrated with the fiberglass waterproof system.

Two primary types of railing posts are commonly encountered in fiberglass deck construction and repair:

- **4x4 pressure-treated railing posts**
- **Structural mount posts used with modern railing systems**

Although these two systems serve the same purpose of supporting the railing structure, their installation methods and waterproofing requirements differ.

4x4 Pressure-Treated Railing Posts

Traditional railing systems often use **4x4 pressure-treated wood posts** that are anchored directly to the deck framing.

These posts provide strong structural support and are frequently covered with decorative vinyl sleeves that match the railing system.

When properly installed with adequate structural blocking between joists, 4x4 posts provide excellent long-term stability.

Because the fiberglass deck membrane surrounds the base of the post, proper flashing must be installed to maintain the waterproof integrity of the deck.

Structural Mount Posts

Modern railing systems frequently use **structural mount posts** that are attached directly to the deck surface.

These posts are typically fastened through the deck using structural screws and are later covered with decorative vinyl sleeves.

Although structural mount posts are widely used in residential railing systems, they introduce multiple fastener penetrations through the waterproof deck surface.

If these penetrations are not properly sealed and flashed, water can infiltrate through the fastener holes and begin damaging the plywood substrate beneath the fiberglass membrane.

It is important to note that structural mount posts themselves are not inherently problematic. When installed correctly with proper waterproof flashing, they can function effectively within fiberglass deck systems.

However, improper installation without proper sealing is a common cause of deck leaks.

Waterproof Flashing Around Posts

Regardless of the type of railing post used, all deck penetrations must be properly waterproofed.

The recommended method for sealing railing post penetrations in fiberglass decks is the installation of fiberglass flashing around the base of each post.

Fiberglass flashing typically extends approximately:

6 inches vertically around the post

This flashing creates a continuous waterproof seal between the post and the fiberglass membrane.

Proper flashing ensures that water cannot enter the deck system through the post penetration.

Structural Blocking for Post Installation

Proper structural support beneath railing posts is essential for both safety and durability.

When installing or repairing railing posts, blocking is often installed between joists to provide a solid structural base for the post.

Blocking pieces are typically constructed using lumber of the same width as the deck joists.

For example:

- decks framed with 2x10 joists use 2x10 blocking
- decks framed with 2x12 joists use 2x12 blocking

Installing blocking of matching size improves load distribution and ensures that post fasteners are anchored securely into structural wood members.

Repair of Damaged Railing Post Areas

In cases where water infiltration has occurred around railing posts, localized repairs may be necessary.

Common signs of damage include:

• soft spots near railing posts
• cracked fiberglass around post bases
• loose or unstable railing posts

Repair procedures typically involve:

1. Removing the existing railing system or post.
2. Cutting out damaged plywood around the post location.
3. Inspecting the structural framing beneath the plywood.
4. Installing structural blocking if necessary.
5. Replacing damaged plywood with new plywood of the same thickness.
6. Reinstalling the post system.
7. Installing fiberglass flashing around the base of the post.
8. Reapplying fiberglass and gelcoat to restore the waterproof membrane.

Properly repairing railing post penetrations helps restore both structural stability and waterproof integrity.

Converting Structural Mount Posts to 4x4 Posts

In some repair situations, structural mount posts may be replaced with traditional **4x4 pressure-treated posts**.

This conversion may be recommended when the original mounting system has caused repeated waterproofing problems or when additional structural stability is desired.

Because 4x4 posts are integrated more directly into the deck framing, they can provide improved structural support when properly installed with blocking and fiberglass flashing.

Integration with Vinyl Sleeve Railing Systems

Many modern railing systems use decorative vinyl sleeves that slide over the structural posts.

These sleeves serve purely aesthetic purposes and do not provide structural support.

The structural integrity of the railing system depends entirely on the post installation beneath the sleeve.

For this reason, all waterproofing and structural reinforcement must be completed before the vinyl sleeve and railing components are installed.

Proper coordination between fiberglass installation and railing installation helps ensure long-term performance of the deck system.

Chapter 10

Door Removal and Waterproof Door Transition Systems

Door transitions are among the most critical waterproofing details in fiberglass deck construction. Because doors connect interior living spaces directly to the exterior deck surface, any failure in waterproofing at the door threshold can allow water to enter the building structure.

Improper door flashing is a common source of water infiltration in fiberglass deck systems. If water penetrates beneath the door frame, it may travel into the wall assembly, damage the deck substrate, or infiltrate the interior structure of the building.

Correct installation and repair of door flashing systems is essential for maintaining the waterproof integrity of both the deck and the building envelope.

Common Causes of Door-Related Leaks

Door leaks may develop when the transition between the fiberglass deck membrane and the door frame is not properly sealed.

Common causes include:

• missing fiberglass flashing beneath the door frame
• improper integration between deck waterproofing and wall waterproofing systems
• deteriorated framing beneath the door threshold
• gaps between door components and surrounding trim
• poor installation of flashing tapes or housewrap connections

Because door assemblies involve multiple building materials and layers, waterproof integration must be carefully executed.

Situations Requiring Door Removal

In many cases, proper repair of door flashing requires temporary removal of the door assembly.

Door removal may be necessary when:

• water infiltration has damaged the framing beneath the threshold
• flashing was never installed beneath the door frame
• previous waterproofing repairs were improperly performed
• fiberglass flashing cannot be installed without removing the door

Removing the door allows technicians to access the underlying framing and install proper waterproofing components.

Door Removal Procedure

Door removal procedures typically involve several steps to allow safe access to the underlying structure.

These steps may include:

1. Removing interior trim around the door frame.

2. Removing exterior trim components surrounding the door.
3. Carefully removing the door unit from the wall opening.
4. Inspecting the framing and plywood beneath the door threshold.

Once the door assembly is removed, technicians can evaluate the condition of the underlying materials.

Repairing the Door Base Structure

If water infiltration has occurred beneath the door, the plywood or framing beneath the threshold may have deteriorated.

Repair procedures may include:

• removing damaged plywood sections
• replacing deteriorated framing components
• rebuilding the structural base beneath the door

In some situations, the base framing beneath the door may need to be slightly raised or corrected to prevent future water infiltration.

Ensuring that the door base sits above the finished fiberglass deck surface helps reduce the risk of water entering the building.

Installation of Fiberglass Door Flashing

Once the structural base beneath the door has been repaired, fiberglass flashing is installed to create a waterproof transition between the deck and the door opening.

The fiberglass flashing integrates the deck membrane with the wall assembly surrounding the door.

Proper fiberglass flashing ensures that water flowing across the deck surface cannot penetrate beneath the door frame.

The flashing must be carefully integrated with the surrounding wall waterproofing system.

Integration with Housewrap and Flashing Tapes

In addition to fiberglass flashing, the door transition must also be integrated with the building's wall waterproofing system.

This typically involves installing:

• flashing tapes around the door frame
• integration with housewrap or siding paper
• sealing connections between flashing layers

Proper layering ensures that water is always directed outward and away from the building structure.

When correctly installed, these components create a continuous waterproof barrier between the deck surface and the building envelope.

Reinstallation of the Door Assembly

After all waterproofing and structural repairs have been completed, the door assembly can be reinstalled.

The reinstallation process typically includes:

• reinstalling the door frame within the wall opening
• applying spray foam insulation around the door perimeter
• installing exterior PVC trim components
• reinstalling interior trim elements

Exterior PVC trim provides a durable, rot-resistant finish around the door frame.

Interior trim components are typically sealed using silicone caulking to prevent air infiltration.

Importance of Proper Door Waterproofing

Door transitions represent a complex intersection of building systems.

If not properly waterproofed, water infiltration at this location can damage both the deck system and the interior structure of the building.

Proper flashing installation, careful integration with wall waterproofing systems, and correct structural preparation beneath the door are essential for long-term performance.

Routine inspection of door transitions can help identify early signs of deterioration and prevent costly structural damage.

Chapter 11

Wall Caps and Parapet Waterproof Systems

In some deck designs, the perimeter of the deck is enclosed by vertical walls or parapets rather than open railings. These enclosed deck configurations require specialized waterproofing systems to protect the top surface of the wall and prevent water from infiltrating the building structure.

Wall caps and parapet tops are particularly vulnerable to water infiltration because they are horizontal surfaces exposed to rain and environmental conditions.

Without proper waterproofing, water can penetrate through joints and seams and eventually enter the wall assembly below.

For this reason, wall caps must be treated as part of the overall fiberglass waterproofing system.

Parapet Wall Conditions

Parapet walls are commonly found in decks where the perimeter is completely enclosed by walls rather than open railings.

These walls may form part of the building structure or may be constructed as protective barriers around rooftop decks.

Because enclosed decks do not allow water to drain off the edge like open decks, these systems typically rely on:

- scuppers
- drains
- internal drainage channels

The architectural design normally defines the drainage locations before the fiberglass waterproofing system is installed.

Proper slope toward these drainage points is essential to prevent standing water on the deck surface.

Importance of Wall Cap Waterproofing

The top surface of a wall or parapet acts as a horizontal surface that collects rainwater.

If the wall cap is not properly waterproofed, water can penetrate through:

• joints in the wall cap surface
• fastener penetrations
• cracks in the cap material

Once water enters the wall cap assembly, it may travel downward into the wall structure and cause hidden damage.

Installing fiberglass waterproofing on the wall cap provides a seamless protective barrier that prevents water from entering the wall system.

Wall Cap Construction in New Installations

In new construction projects, wall caps are typically constructed using a layered system similar to the deck surface.

The process may include the following steps:

1. Installing **5/4 pressure-treated packer boards** on both sides of the parapet wall.
2. Installing **1/2 inch ACX plywood** across the top of the wall to form the cap surface.
3. Securing the plywood using glue and screws to create a stable substrate.

This construction method creates a rigid and stable base for the fiberglass waterproof membrane.

Drip Edge Installation on Wall Caps

After the plywood cap surface is installed, a **PVC drip edge** is typically installed along the edges of the wall cap.

PVC drip edge is commonly used because it is:

• rot resistant
• moisture resistant
• durable in coastal environments

The drip edge helps direct water away from the wall face and prevents water from running down behind exterior siding or wall finishes.

Fiberglass Installation on Wall Caps

Once the cap surface is prepared, fiberglass reinforcement is installed over the entire wall cap surface.

The fiberglass membrane typically includes:

• fiberglass mat reinforcement
• resin saturation
• full coverage across the wall cap surface

The fiberglass layer creates a continuous waterproof membrane that protects the wall cap from moisture infiltration.

The fiberglass may also extend slightly down the sides of the parapet to ensure full coverage and proper waterproofing at the transitions.

Surface Preparation and Gelcoat Finish

After the fiberglass layer has cured, the surface is ground to remove irregularities and prepare the surface for finishing.

The wall cap is then cleaned with acetone to remove dust and contaminants.

Finally, a gelcoat finish is applied to protect the fiberglass surface and provide a durable exterior finish.

The gelcoat layer also protects the fiberglass reinforcement from UV exposure and environmental wear.

Wall Cap Repair Procedures

In existing buildings, wall caps may require repair due to age, water infiltration, or previous installation failures.

Repair procedures often involve removing sections of roofing materials where the roof meets the parapet wall.

This allows technicians to install proper flashing systems beneath the roofing layers.

Typical wall cap repair procedures may include:

- removing roofing materials near the parapet
- installing packer boards and plywood cap structure
- installing PVC drip edge
- installing fiberglass flashing where the roof meets the wall
- reinstalling roofing materials over the flashing system

This layered approach ensures that the roof and deck waterproofing systems work together to prevent water infiltration.

Integration Between Roof and Deck Waterproofing

When parapet walls connect roof surfaces with fiberglass decks, proper integration between roofing materials and fiberglass flashing becomes essential.

Roofing materials must always overlap the flashing layers in a way that directs water away from the wall assembly.

This principle of layered waterproofing ensures that water always flows outward rather than entering the building structure.

Failure to maintain this layering sequence can result in hidden leaks that may damage both the deck and the roof structure.

Long-Term Durability of Wall Cap Systems

Wall caps are constantly exposed to weather conditions, including:

• rain
• UV exposure
• temperature changes
• coastal humidity

Because of this exposure, durable materials such as fiberglass membranes and PVC components provide long-term protection against moisture damage.

When properly installed and maintained, fiberglass wall cap systems can significantly extend the lifespan of parapet wall assemblies.

Chapter 12

Demolition, Preparation, and Site Control Procedures

Before any fiberglass deck installation, restoration, or structural repair can begin, the job site must be properly prepared. In many projects, this preparation includes the controlled demolition of existing deck materials, removal of deteriorated components, and preparation of the structural framing for new construction.

Proper demolition and site preparation are essential to ensure that the new fiberglass system is installed on a stable and clean substrate.

Careful handling of demolition work also helps prevent unnecessary damage to structural components that must remain in place.

Controlled Deck Demolition

In many repair and reconstruction projects, existing deck surfaces must be removed before the new system can be installed.

Typical deck demolition may involve removing:

• existing fiberglass membranes
• gelcoat finishes
• deteriorated plywood substrate layers

- damaged drip edges
- other deteriorated deck components

Demolition work must be performed carefully to avoid damaging the structural framing that supports the deck.

Whenever possible, demolition should be limited to the components that require replacement.

Protection of Structural Framing

During demolition, the underlying structural framing must be protected.

Joists, beams, and structural posts should not be damaged during removal of deck materials.

Careful cutting and removal techniques should be used to separate plywood panels and fiberglass layers from the framing members.

If structural components are accidentally damaged during demolition, additional repairs may be required before reconstruction can proceed.

Removal of Temporary Components

Certain deck components may need to be temporarily removed before demolition and reconstruction can begin.

These components may include:

- railing systems
- railing posts
- exterior trim elements
- fascia boards
- soffit or ceiling panels beneath the deck

Temporary removal of these components allows technicians to access the deck structure and perform repairs safely.

After the deck work is complete, these components can be reinstalled or replaced as needed.

Site Debris Management

Demolition of deck systems can produce a large amount of debris, including wood materials, fiberglass fragments, and other construction waste.

Proper debris management is important for maintaining a safe and organized work environment.

Typical debris management procedures include:

• collecting demolition materials in heavy-duty bags or containers
• separating recyclable materials when possible
• removing debris from the job site regularly

Maintaining a clean job site reduces safety hazards and improves overall work efficiency.

Preparation of Structural Surfaces

Once demolition is complete, the structural deck framing must be carefully inspected and prepared for reconstruction.

Technicians should verify:

• structural integrity of joists and beams
• proper alignment of framing members
• absence of rot or moisture damage

If any structural defects are identified, repairs must be completed before installing new plywood or fiberglass materials.

Ensuring a stable structural base is essential for long-term performance of the fiberglass deck system.

Cleaning and Surface Preparation

Before installing new plywood layers or fiberglass materials, the work area must be thoroughly cleaned.

Dust, debris, and loose materials should be removed from the deck structure.

Cleaning the framing and work surfaces helps ensure proper installation of new components and prevents contaminants from interfering with bonding processes.

Disposal of Demolition Materials

Construction debris must be disposed of in accordance with local waste management regulations.

Demolition materials are typically transported to approved disposal facilities or construction debris dumps.

Proper disposal ensures that the project complies with environmental regulations and prevents illegal dumping of construction waste.

Job Site Safety Considerations

Deck demolition and reconstruction work involves several potential safety hazards.

Common safety considerations include:

- safe use of power tools
- proper handling of fiberglass materials
- protection from airborne dust and debris
- safe access to elevated work areas

Following proper safety procedures helps protect workers and ensures that projects are completed without accidents or injuries.

Importance of Proper Preparation

Successful fiberglass deck installation depends heavily on proper preparation of the job site.

By carefully controlling demolition work, protecting structural components, and preparing the work environment, contractors can ensure that the new fiberglass system is installed under optimal conditions.

Thorough preparation reduces the likelihood of installation errors and contributes to the long-term durability of the fiberglass waterproof deck system.

SEASHORE FIBERGLASS LLC

MASTER TECHNICAL MANUAL

Volume 2 — Inspection and Diagnostics

Chapter 1

Professional Fiberglass Deck Inspection Method

Proper inspection of fiberglass waterproof decks is essential for identifying early signs of deterioration and preventing structural damage. Many deck failures begin with small issues that, if left unresolved, can eventually compromise the entire deck system.

A systematic inspection method allows technicians to evaluate the condition of the fiberglass membrane, the plywood substrate, and critical waterproofing transitions.

Professional deck inspections should follow a logical sequence that begins with visual observation and continues with physical testing and detailed investigation of potential problem areas.

Initial Visual Inspection

The first step in evaluating a fiberglass deck is a visual inspection of the entire surface and perimeter.

Technicians should carefully observe the deck surface for visible signs of deterioration.

Common visual indicators include:

- large cracks in the fiberglass surface
- clusters of small cracks (spider cracking)
- worn or faded gelcoat
- surface discoloration caused by moisture
- deteriorated or damaged drip edges

These visible signs may indicate aging fiberglass systems or underlying structural movement.

While visual inspection provides important information, many underlying problems may not be immediately visible.

Interpreting Surface Cracks

Fiberglass decks may develop cracks over time due to environmental exposure, substrate movement, or aging of the fiberglass system.

When technicians observe large cracks or widespread cracking patterns, this often indicates that the fiberglass membrane has reached the end of its service life.

In many cases, widespread cracking suggests that the deck may require a **full reglass restoration** rather than a simple recolor.

Large cracks also present a risk of water infiltration if they remain unrepaired.

Water entering through surface cracks can reach the plywood substrate and eventually cause structural deterioration.

Walking Inspection for Soft Spots

After completing the visual inspection, technicians should perform a walking inspection across the entire deck surface.

Walking the deck allows technicians to detect structural weaknesses beneath the fiberglass membrane.

Areas that feel:

- soft
- spongy
- unstable under foot pressure

may indicate deterioration of the plywood substrate.

Soft spots are often easiest to detect when the technician shifts body weight while walking across the deck.

In severe cases, soft spot areas may also appear visibly depressed.

Common Locations for Soft Spot Development

Soft spots can occur in many areas of the deck, but certain locations are particularly susceptible to water infiltration.

Technicians should pay special attention to the following areas:

- near walls connected to the deck
- near door transitions
- around railing posts
- around structural posts inside decorative columns
- along deck edges and perimeter areas

All of these locations involve waterproof transitions that can fail if flashing systems are improperly installed.

Water Flow and Leak Patterns

Understanding how water flows across the deck surface can help technicians determine the origin of leaks.

In many decks, walls and doors are located at the highest elevation of the deck surface.

If water infiltration begins at these upper transition points, water may travel downward beneath the fiberglass membrane.

As a result, damage may spread across large portions of the deck surface.

In these cases, the original leak source may be located near the upper section of the deck even if soft spots appear in lower areas.

Diagnosing Flashing Failures

Flashing problems are one of the most common causes of fiberglass deck leaks.

Technicians should carefully inspect areas where the deck surface meets vertical structures such as walls and doors.

Signs that flashing problems may be present include:

• plywood that feels soft near the wall
• concentrated cracking patterns along wall transitions
• signs of water infiltration near door thresholds

When these symptoms appear along the same line as a wall or door, missing or improperly installed fiberglass flashing is often the underlying cause.

In many cases, siding may need to be temporarily removed to inspect and repair the flashing system.

Drip Edge Inspection

The perimeter edge of the deck should also be carefully inspected.

One of the most common problems found during deck inspections is deterioration of wooden drip edges.

Older decks were sometimes constructed using wood components at the deck edge, which can deteriorate after long-term exposure to moisture.

Signs of drip edge failure include:

• rotted wood at the deck edge
• separation between the deck surface and the edge material
• improper water drainage at the perimeter

Replacing deteriorated drip edges with PVC or Azek materials can significantly improve long-term durability.

Inspection of Railing Posts

Railing posts are another critical inspection point.

Both traditional **4x4 posts** and **structural mount posts** can become sources of water infiltration if not properly waterproofed.

Technicians should inspect posts for:

• cracks in the fiberglass surrounding the post
• broken or deteriorated gelcoat around the base
• instability or movement in the railing system

Movement in railing posts can indicate structural issues beneath the deck surface or failed waterproof penetrations.

If cracks or instability are observed, further investigation of the post installation may be required.

Service Life Expectations

Fiberglass decks are durable systems, but they require periodic maintenance.

In coastal environments such as the Jersey Shore, typical maintenance intervals may include:

Recoloring:
approximately every **5 years**

Recoloring restores the protective gelcoat layer and maintains the appearance of the deck surface.

Reglass restoration:
approximately every **10 years**

Reglassing installs a new fiberglass reinforcement layer to renew the waterproof membrane.

Actual maintenance intervals may vary depending on:

- environmental exposure
- quality of the original installation
- level of ongoing maintenance

Regular inspection helps identify issues before major repairs become necessary.

Chapter 2

Leak Source Identification in Fiberglass Deck Systems

Correctly identifying the origin of a leak is one of the most important steps in fiberglass deck repair. Water infiltration may originate from multiple sources, including the fiberglass membrane itself, transition flashings, railing posts, doors, or structural columns.

Because water can travel beneath the fiberglass membrane before appearing as visible damage, technicians must carefully analyze damage patterns across the deck surface to determine the true source of the leak.

Proper diagnosis prevents unnecessary repairs and ensures that the underlying problem is corrected.

Leaks Originating from the Fiberglass Surface

Although transition failures are a common cause of leaks, water can also penetrate directly through the fiberglass field when the membrane becomes severely damaged.

Indicators that a leak may originate directly from the fiberglass surface include:

- large cracks in the fiberglass membrane
- areas where the gelcoat has completely worn away
- exposed plywood where fiberglass has deteriorated
- visible separation or failure of the fiberglass layer

When the fiberglass membrane has deteriorated to this level, water can easily penetrate the deck system.

In these cases, localized repairs may not be sufficient, and a **full reglass restoration** is often required.

Leaks Originating from Wall Flashings

Wall transitions are one of the most common sources of water infiltration in fiberglass decks.

When flashing between the deck and the wall is missing or improperly installed, water can enter beneath the fiberglass membrane and travel downward through the deck assembly.

Typical warning signs include:

- clusters of cracks along the wall line
- soft plywood near the wall transition
- cracking patterns that follow the length of the wall

These symptoms often indicate that water is entering at the wall transition rather than through the fiberglass field.

In many cases, siding must be temporarily removed to inspect and repair the flashing system.

Leaks Originating from Door Transitions

Door thresholds represent another critical transition point where leaks frequently develop.

Water infiltration around doors often begins directly in front of the door opening and then spreads across the deck surface.

A common damage pattern includes:

• cracking or deterioration in the deck surface directly in front of the door
• soft spots forming near the threshold
• damage spreading downward across the deck as water travels beneath the fiberglass membrane

Because doors are typically located at the highest point of the deck, leaks at this location can allow water to travel across large areas of the deck before being detected.

Repairing door leaks often requires removal of the door assembly in order to install proper flashing beneath the threshold.

Leaks Originating from Railing Posts

Railing posts are another frequent source of water infiltration.

Posts that penetrate the fiberglass membrane must be properly flashed to prevent water from entering through fastener penetrations.

Early warning signs of railing post leaks include:

• cracks in the fiberglass surrounding the base of the post
• soft spots forming around the post location
• instability or looseness in the railing system
• broken or deteriorated gelcoat around the post

If these symptoms are present, the post installation should be carefully inspected for proper flashing and structural support.

Leaks Originating from Structural Columns

Structural posts that are concealed inside decorative columns can also become major leak sources if flashing is missing or improperly installed.

Because these structural posts are often hidden inside column wraps, water infiltration may go unnoticed for long periods of time.

Typical symptoms include:

- soft spots forming around the base of the column
- localized deck deterioration near the column location
- damage spreading outward from the column area

If leaks remain uncorrected for extended periods, water infiltration around structural columns may eventually damage the structural joists supporting the deck.

When column-related leaks are suspected, the base trim of the column may need to be removed in order to inspect the flashing around the structural post.

Identifying When a Deck Requires Reglass

During inspection, technicians must determine whether the deck requires localized repair, recolor maintenance, or a full reglass restoration.

A deck typically requires reglass when the fiberglass membrane itself has reached the end of its service life.

Common indicators that reglass may be required include:

- widespread cracking across the entire deck surface
- aged or deteriorated fiberglass reinforcement
- multiple previous repair patches across the deck
- deteriorated or rotted drip edge components

When several of these conditions are present, installing a new fiberglass layer across the entire deck is often the most reliable restoration method.

Identifying When Recolor Maintenance Is Appropriate

In many cases, the fiberglass membrane itself remains structurally sound while the protective gelcoat surface becomes worn.

When this occurs, the deck may only require recolor maintenance rather than full reglass restoration.

Typical signs that recolor maintenance is appropriate include:

- faded or worn gelcoat finish
- minor surface wear without structural fiberglass damage
- small cosmetic cracks that have not penetrated the fiberglass membrane

Recolor maintenance restores the protective gelcoat layer and helps extend the lifespan of the fiberglass deck system.

Importance of Accurate Leak Diagnosis

Accurate leak diagnosis is critical for selecting the correct repair strategy.

Misidentifying the source of water infiltration can result in incomplete repairs that allow leaks to return.

By carefully analyzing surface patterns, structural conditions, and transition details, technicians can determine the true origin of deck leaks and implement the most effective repair solution.

Chapter 3

Water Travel Patterns in Fiberglass Deck Systems

One of the most important aspects of fiberglass deck diagnostics is understanding how water travels beneath the fiberglass membrane after infiltration occurs.

In many cases, the visible damage on a deck surface does not represent the original location where water entered the system. Because fiberglass decks are installed with a slope to allow water drainage, infiltrated water often travels beneath the membrane following the natural pitch of the deck.

This movement can cause damage to appear far from the original leak source.

Understanding these water travel patterns allows technicians to properly diagnose leaks and identify the true origin of deck failures.

Water Movement Beneath the Fiberglass Membrane

When water enters a fiberglass deck system through cracks, failed flashings, or penetrations, it may reach the plywood substrate beneath the fiberglass membrane.

Once water reaches the substrate, it often begins moving along the surface of the plywood.

Because fiberglass decks are typically constructed with a slope of approximately:

1/4 inch per foot

water beneath the membrane tends to follow this slope.

As a result, water infiltration rarely remains confined to the original entry point.

Instead, the water gradually moves downward across the deck structure following the direction of the deck pitch.

Leak Sources at Higher Deck Elevations

In many residential deck designs, walls and door transitions are located at the higher elevation of the deck surface.

When flashing failures occur at these locations, water may enter beneath the fiberglass membrane at the upper section of the deck.

From there, water travels downward beneath the membrane following the deck slope.

This movement can allow water to spread across large sections of the deck before visible damage appears.

Spread of Damage Over Time

When water infiltration is not repaired promptly, moisture can continue spreading beneath the fiberglass membrane for extended periods.

As the water moves along the deck slope, it may gradually affect larger areas of the plywood substrate.

Over time, this can result in:

- multiple soft spots appearing across the deck
- structural weakening of plywood panels
- deterioration of structural framing beneath the deck

In severe cases, water infiltration that begins at the top of the deck may eventually reach the lowest edge of the deck structure.

If left unresolved for long periods, the damage may extend into structural joists.

Identifying the True Leak Location

Because water can travel significant distances beneath the fiberglass membrane, technicians must carefully evaluate the entire deck system when diagnosing leaks.

Visible damage such as soft spots may appear in lower areas of the deck even when the original leak source is located at a higher elevation.

For this reason, the location of visible damage does not always indicate the true source of water infiltration.

Initial Investigation of Soft Spot Areas

When technicians identify a soft spot during inspection, the first step is to investigate the damaged area directly.

The affected section of the deck should be carefully evaluated to determine the extent of the substrate deterioration.

Inspection of the immediate area may reveal clues such as:

- localized fiberglass cracks
- deteriorated plywood
- compromised post penetrations

These indicators may help determine whether the damage originated locally or resulted from water traveling beneath the membrane from another location.

Importance of Early Repairs

Because water can travel long distances beneath the fiberglass membrane, early detection and repair of leaks is critical.

Small leaks that remain unrepaired can eventually lead to widespread structural damage across the entire deck surface.

Prompt maintenance and repair help limit the spread of moisture and reduce the extent of future repairs.

Routine inspections allow technicians to identify early warning signs before significant structural deterioration occurs.

Chapter 4

Advanced Inspection Techniques for Fiberglass Deck Systems

Professional inspection of fiberglass decks requires more than simply identifying visible damage. Experienced technicians are often able to detect early warning signs of failure before structural deterioration becomes severe.

Recognizing these early indicators allows contractors to perform preventative repairs and avoid extensive structural damage.

Advanced inspection techniques focus on identifying subtle patterns in cracking, structural movement, and waterproofing transitions.

Early Indicators of Deck Failure

Before soft spots develop, fiberglass decks often display early warning signs that indicate underlying movement or water infiltration.

Common early indicators include:

• clusters of small cracks concentrated in specific areas
• cracking patterns that follow a straight line across the deck surface
• cracking around railing posts or structural penetrations
• localized cracking near walls or transitions

These cracking patterns often indicate movement in the plywood substrate beneath the fiberglass membrane.

When multiple cracks appear in a concentrated area, the technician should investigate the possibility of underlying moisture damage.

Inspection Around Decorative Columns

Decorative columns commonly conceal structural posts that support the deck or roof structures above.

Because the structural components inside these columns are hidden from view, leaks and flashing failures may remain undetected for long periods.

Inspection around columns typically begins by carefully walking around the base of the column.

Technicians apply pressure around the column perimeter to detect:

• soft areas in the deck surface
• localized instability
• visible cracking around the column base

Cracks surrounding the column area often indicate potential flashing failures around the structural post.

In many decks, railing systems are attached directly to the column assemblies. Because of this, railing components may need to be removed before the column trim can be disassembled for deeper inspection.

Lower trim moldings around the column base are typically removed during repair work to inspect the flashing condition around the structural post.

Inspecting Railing Posts with Vinyl Sleeves

Modern railing systems often use decorative vinyl sleeves that conceal the structural posts beneath them.

Because the structural post is hidden inside the sleeve, visual inspection of the post base becomes more difficult.

Technicians must rely on indirect indicators to detect problems.

Common warning signs include:

- movement or instability in the railing system
- cracking in the fiberglass surrounding the base of the post
- broken or deteriorated gelcoat around the post penetration

Movement in the railing system may indicate structural instability or failed waterproofing around the post penetration.

If these symptoms are observed, further investigation of the post installation may be required.

Diagnosing Hidden Wall Flashing Problems

In many existing homes, siding is installed over the wall transition where fiberglass flashing should be located.

Because the flashing is hidden behind the siding, technicians must rely on surface symptoms to identify potential problems.

Warning signs that flashing may be missing or improperly installed include:

- cracking patterns that follow the line of the wall
- soft spots located directly along the wall transition
- water infiltration symptoms appearing inside the building

When these indicators appear together, the technician may suspect that fiberglass flashing beneath the siding is missing or improperly installed.

In such cases, siding may need to be temporarily removed to inspect and repair the flashing system.

Diagnosing Drip Edge Problems

The deck edge is another critical inspection point where waterproofing failures can occur.

In many older decks, wooden drip edges were used instead of modern PVC materials.

In some construction scenarios, exterior fascia boards are installed flush with the drip edge. This situation often occurs when a packer board was not installed beneath the plywood during deck construction.

When the fascia and drip edge share the same thickness, the fascia may sit flush with the drip edge rather than fitting beneath it.

This condition prevents proper water drainage and allows water to infiltrate between the drip edge and the fascia board.

Over time, this infiltration can cause moisture to accumulate behind the fascia, potentially leading to deterioration of the structural joists hidden behind the fascia.

Correcting Improper Drip Edge Systems

When drip edge systems fail due to improper construction, corrective measures may involve rebuilding the edge assembly.

If the structural framing beneath the deck edge has not been damaged, the repair may involve:

- removing the existing wooden drip edge
- installing a packer board of equal thickness to the removed drip edge
- installing a larger PVC drip edge, such as a 1x6 PVC board

The new drip edge extends beyond the fascia and directs water properly over the fascia surface.

This configuration restores the correct drainage path and prevents water from infiltrating behind the fascia boards.

Identifying Structural Joist Damage

When leaks remain unresolved for extended periods, moisture infiltration may eventually affect the structural joists supporting the deck.

Signs that structural damage may be present include:

• multiple soft spots across the deck surface
• noticeable sagging or deformation of the deck structure
• instability when walking across the deck
• visible structural distortion of the framing

In severe cases, the deck may become unsafe to use.

Structural joist deterioration caused by prolonged water infiltration can compromise the load-bearing capacity of the deck.

If these symptoms are present, the deck should be evaluated carefully before allowing continued use.

Importance of Professional Inspection

Advanced inspection techniques allow experienced technicians to detect problems early and identify hidden failures within fiberglass deck systems.

By recognizing subtle warning signs and understanding the structural behavior of waterproof deck systems, contractors can diagnose problems more accurately and recommend the most appropriate repair strategy.

Early detection and correct diagnosis help prevent major structural damage and extend the lifespan of fiberglass deck systems.

Chapter 5

Common Fiberglass Deck Failure Scenarios

Fiberglass deck systems are highly durable when properly installed and maintained. However, many decks eventually develop problems due to installation errors, aging materials, or lack of maintenance.

In coastal regions such as the Jersey Shore, fiberglass decks are exposed to heavy moisture, salt air, and seasonal temperature changes. These environmental factors can accelerate deterioration when waterproofing systems fail.

Through field experience, several common failure scenarios appear repeatedly in fiberglass deck systems.

Understanding these patterns helps technicians quickly identify the root cause of leaks and structural deterioration.

Multiple Failure Points in Fiberglass Decks

Unlike some construction systems where a single defect may dominate, fiberglass decks often develop problems from multiple sources simultaneously.

Several components of the deck system are potential leak points, including:

- wall flashings
- door flashings
- railing post penetrations
- drip edge systems
- aging fiberglass membranes

Because all of these components work together to maintain waterproofing, failure in any one area can eventually lead to water infiltration.

In many cases, technicians may find several of these issues present in the same deck.

Failure Scenario: Structural Post Inside Decorative Columns

One of the most difficult leak sources to detect occurs around structural posts that are concealed inside decorative column assemblies.

These columns are often constructed using PVC or composite materials and are installed primarily for aesthetic purposes. The structural support post is hidden inside the column wrap.

If fiberglass flashing around the structural post is missing or improperly installed, water may infiltrate the deck system around the base of the column.

Because the structural post is concealed, early detection is difficult.

In many cases, homeowners only become aware of the problem when the damage has already progressed significantly.

Common signs of this failure include:

• soft spots forming around the column base
• visible cracking in the deck surface near the column
• structural deterioration beneath the column area

If water infiltration continues for long periods, moisture may eventually reach the structural joists supporting the deck.

Failure Scenario: Installation Errors in Newer Decks

Even relatively new fiberglass decks may develop problems if installation details were not executed correctly.

Common installation errors found in decks that are five to ten years old include:

• railing posts that were not properly sealed or flashed
• wall flashings that were improperly installed
• drip edge systems that were installed incorrectly

These errors often result from improper sequencing during construction or lack of familiarity with fiberglass waterproofing systems.

Because fiberglass decks depend heavily on proper flashing and transition details, small installation mistakes can eventually lead to leaks.

Failure Scenario: Aging Deck Systems

Older fiberglass decks that have been exposed to weather conditions for many years often develop multiple forms of deterioration simultaneously.

Common issues found in decks that are fifteen to twenty years old include:

• severely aged fiberglass membranes
• deteriorated wooden drip edges
• numerous previous repair patches
• degraded gelcoat surfaces

When several of these conditions are present, the deck system may require extensive restoration or complete reconstruction.

Severe Structural Failure

In the most advanced failure scenarios, prolonged water infiltration can lead to structural deterioration of the deck framing itself.

Severe structural failure may involve:

• completely deteriorated plywood substrate
• rotted structural joists
• instability across the deck surface
• visible structural deformation

At this stage, the deck may become unsafe for normal use.

Walking across the deck may produce noticeable movement or instability, indicating that the structural framing has lost significant strength.

When structural deterioration reaches this level, the deck typically requires major structural reconstruction rather than simple surface repairs.

Importance of Early Intervention

Many of the severe failure scenarios described above develop slowly over time.

Small leaks that go unnoticed for years can eventually lead to widespread damage across the deck structure.

Routine inspection and timely repairs allow technicians to address problems early, before they develop into structural failures.

Early intervention significantly reduces repair costs and helps extend the lifespan of fiberglass waterproof deck systems.

SEASHORE FIBERGLASS LLC

MASTER TECHNICAL MANUAL

Volume 3 — Repair Systems

Chapter 1

Professional Reglass System for Fiberglass Deck Restoration

Reglass restoration is one of the most effective methods for renewing the waterproof membrane of an existing fiberglass deck. When performed correctly, reglassing installs a completely new fiberglass layer over the existing deck system, restoring full waterproof protection.

This process is typically used when the fiberglass membrane has deteriorated due to aging, widespread cracking, or repeated previous repairs.

Proper preparation is essential to ensure that the new fiberglass membrane bonds correctly to the existing deck surface and that underlying structural components remain stable.

Step 1 — Removal of Railing Systems

The first step in a professional reglass project is the removal of railing systems.

Temporary removal of railing components allows technicians to access the full deck surface, including perimeter areas that would otherwise be obstructed.

Railing posts, rails, and decorative components are carefully removed and stored so they can be reinstalled after the deck restoration is completed.

Removing these components ensures that the fiberglass membrane can be installed continuously across the entire deck surface without interruptions.

Step 2 — Structural Inspection

After the railing system has been removed, the deck surface and perimeter areas are inspected for structural issues.

Technicians examine:

- the plywood substrate
- drip edge components
- flashing transitions
- post penetrations
- column bases

Any necessary repairs must be completed before installing the new fiberglass membrane.

Common repairs performed during this stage include replacing deteriorated drip edges, correcting flashing issues, and repairing localized soft spots.

Step 3 — Re-Securing the Plywood Substrate

Older fiberglass decks were often constructed using nails to secure the plywood to the framing.

Over time, nail fasteners may loosen, allowing slight movement in the plywood substrate. This movement can lead to cracking in the fiberglass membrane.

To stabilize the deck surface, the plywood is re-secured by installing screws into the framing.

Screws are typically installed:

- along each joist
- across the entire deck surface
- spaced approximately **6 inches apart**

This step significantly increases the structural stability of the deck surface and helps prevent future fiberglass cracking.

Although this process penetrates the existing fiberglass layer, the new fiberglass membrane installed during reglassing will fully restore the waterproof barrier.

Step 4 — Drip Edge and Structural Repairs

If the deck contains deteriorated drip edge components, they must be replaced before continuing with the fiberglass installation.

Wooden drip edges are often removed and replaced with modern PVC or Azek components that resist moisture and rot.

During this stage, technicians also inspect the plywood and structural framing near the deck edge.

Repairs may include:

• replacing damaged plywood
• repairing edge framing if necessary
• installing new drip edge components

Completing these repairs before fiberglass installation ensures that the deck perimeter remains structurally sound and properly waterproofed.

Step 5 — Surface Grinding

Once all structural repairs have been completed, the entire deck surface is thoroughly ground.

Grinding the deck surface serves several purposes:

• removes contaminants and loose material
• roughens the surface to improve bonding
• smooths transitions between existing fiberglass areas

This step creates the proper mechanical bonding surface required for the new fiberglass layer.

Complete grinding of the deck surface is a critical preparation step.

Step 6 — Filling Screw Heads and Surface Imperfections

After grinding is complete, screw heads and surface irregularities are filled using body filler such as Bondo.

Filling these areas helps eliminate surface imperfections that could telegraph through the fiberglass membrane.

Areas typically filled include:

• screw heads installed during substrate stabilization
• seams between plywood panels
• minor surface cracks or imperfections

Once the filler has cured, the filled areas are smoothed to create an even surface.

Step 7 — Fiberglass Reinforcement Installation

After preparation is complete, the new fiberglass membrane is installed.

The fiberglass system typically consists of:

• **one continuous layer of 2-ounce fiberglass mat**

The fiberglass mat is laid across the entire deck surface and saturated with resin.

Proper saturation ensures that all fiberglass fibers are fully embedded in the resin matrix, forming a strong and continuous waterproof membrane.

The fiberglass layer must be installed continuously across the deck to maintain a seamless waterproof barrier.

Step 8 — Fiberglass Surface Grinding

Once the fiberglass layer has cured, the surface is ground again to prepare it for finishing.

Grinding the cured fiberglass surface removes irregularities and creates a smooth surface for gelcoat adhesion.

This step also ensures that the final finish layer bonds properly to the fiberglass membrane.

Step 9 — Surface Cleaning

Before applying the final finish layer, the deck surface is thoroughly cleaned using acetone.

Cleaning removes dust, oils, and contaminants that could interfere with gelcoat adhesion.

Proper cleaning is essential to ensure that the gelcoat layer bonds correctly to the fiberglass surface.

Step 10 — Gelcoat Application

The final step in the reglass process is the application of gelcoat.

Gelcoat serves as the protective outer layer of the deck system.

Functions of gelcoat include:

- UV protection
- surface durability
- moisture resistance
- slip resistance

Seashore Fiberglass typically applies **textured gelcoat finishes** to provide a safe walking surface.

The textured finish also helps improve traction in wet conditions.

Once the gelcoat cures, the fiberglass deck restoration is complete.

Continuous Deck Restoration

Professional reglass systems restore the waterproof integrity of fiberglass decks and extend their service life.

When performed with proper preparation and installation techniques, reglass restoration creates a durable waterproof membrane that can perform reliably for many years.

Regular maintenance, including periodic recolor applications, helps preserve the gelcoat finish and prolong the life of the fiberglass deck system.

Chapter 2

Professional Recolor System for Fiberglass Deck Maintenance

Recoloring is a maintenance procedure used to restore the protective gelcoat surface of a fiberglass deck without replacing the fiberglass reinforcement layer.

Over time, the gelcoat finish on fiberglass decks may become worn due to exposure to sunlight, foot traffic, and environmental conditions. When the fiberglass membrane remains structurally sound but the gelcoat surface has deteriorated, recolor maintenance can restore both the appearance and protection of the deck.

Proper preparation is essential to ensure that the new gelcoat bonds correctly to the existing fiberglass surface.

Step 1 — Full Deck Surface Grinding

The first step in the recolor process is grinding the entire deck surface.

Grinding removes the worn gelcoat layer and creates a mechanical bonding surface that allows the new gelcoat to adhere properly.

Grinding also reveals hidden surface defects that may not have been visible during initial inspection.

During this step, technicians carefully examine the deck surface for:

- cracks in the fiberglass membrane
- damaged areas in the gelcoat layer
- surface imperfections that require repair

Complete surface grinding is critical to ensure long-term adhesion of the new gelcoat.

Step 2 — Repairing Cracks and Surface Damage

During the grinding process, technicians often discover cracks or damaged areas that must be repaired before applying the new gelcoat.

When cracks are identified, the damaged area is opened and prepared for repair.

Repairs are typically performed by installing a fiberglass patch over the affected area.

The repair process involves:

• preparing the damaged area
• installing fiberglass reinforcement
• saturating the fiberglass with resin

These fiberglass patches restore the structural integrity of the damaged areas before the final finish is applied.

Step 3 — Grinding and Blending the Repair Patches

Once the fiberglass repair patches have cured, they must be ground to blend them smoothly into the surrounding deck surface.

Grinding the patches ensures that the repaired areas become level with the rest of the deck and remain visually unobtrusive after the gelcoat is applied.

Proper blending of repair patches helps create a uniform finished surface.

Step 4 — Surface Cleaning and Dust Removal

After all grinding and repairs are complete, the deck must be thoroughly cleaned before applying the gelcoat.

Cleaning involves removing all dust and debris generated during the grinding process.

Typical cleaning procedures include:

• blowing loose dust from the surface
• vacuuming the entire deck
• wiping the deck surface with acetone

Cleaning the surface with acetone removes any remaining dust, oils, or contaminants that could interfere with gelcoat adhesion.

This step is essential for ensuring proper bonding between the new gelcoat layer and the fiberglass surface.

Step 5 — Gelcoat Application

After the surface has been fully prepared and cleaned, the new gelcoat finish is applied.

Gelcoat serves as the protective outer layer of the fiberglass deck system.

Its functions include:

• protecting the fiberglass reinforcement from UV exposure
• providing a durable walking surface
• maintaining the waterproof integrity of the deck surface

Seashore Fiberglass typically applies **textured gelcoat finishes**, which provide improved traction and reduce the risk of slipping when the deck is wet.

The textured surface also enhances the visual appearance of the finished deck.

Maintenance Cycle

Recolor maintenance is typically recommended every **five years**, depending on environmental exposure and usage.

Routine recolor maintenance helps extend the lifespan of the fiberglass deck system by preserving the protective gelcoat layer.

When properly maintained, fiberglass decks can provide long-term waterproof protection and reliable performance.

Chapter 3

Professional Soft Spot Repair System

Soft spots occur when the plywood substrate beneath the fiberglass membrane becomes weakened due to prolonged moisture exposure. When water infiltrates the deck system and remains trapped beneath the fiberglass layer, the plywood can deteriorate and lose structural strength.

Soft spot repairs focus on removing damaged substrate materials, restoring the structural integrity of the deck, and reinstalling a fully waterproof fiberglass surface.

Proper repair procedures ensure that the damaged area is completely stabilized and that the new fiberglass layer bonds correctly to the existing deck surface.

Step 1 — Identifying the Repair Area

Once a soft spot is detected, the technician must determine the full extent of the damaged plywood.

Soft spots rarely stop exactly at the visible or felt boundary of the damaged area. Moisture may spread beneath the fiberglass membrane and weaken plywood beyond the initial soft area.

For this reason, the repair area is expanded until solid plywood is found.

Technicians typically cut the deck surface until reaching plywood that is completely stable and structurally sound.

This ensures that all deteriorated material is removed before rebuilding the deck surface.

Step 2 — Cutting and Removing Damaged Materials

After defining the repair area, the fiberglass membrane and plywood layers are cut and removed.

Many fiberglass decks contain multiple plywood layers as part of the deck substrate system.

During removal, technicians evaluate the condition of both layers.

If both plywood layers have been affected by moisture damage, both layers must be removed and replaced.

If only one layer is damaged, the repair may involve replacing the affected layer while preserving the remaining structural layer.

Step 3 — Inspecting the Structural Framing

After removing the damaged plywood, the structural framing beneath the deck must be inspected.

Joists and structural supports are evaluated for signs of moisture damage, rot, or structural weakness.

If the framing remains structurally sound, the repair can proceed with plywood replacement.

If structural components have been affected, additional structural repairs may be required before reinstalling the deck surface.

Step 4 — Installing Replacement Plywood

Replacement plywood must match the thickness and configuration of the original deck system.

The new plywood panels are installed using construction adhesive and structural screws.

Fastening typically includes:

- glue between layers
- screws installed along the joists
- screw spacing approximately **6 inches apart**

This fastening method ensures that the new plywood becomes firmly integrated into the existing deck structure and minimizes future movement.

Proper substrate stability is essential for preventing cracks in the fiberglass membrane.

Step 5 — Surface Preparation for Fiberglass Repair

Before installing the fiberglass repair patch, the deck surface must be properly prepared.

Preparation follows the same procedure used for fiberglass installation throughout the deck system.

Surface preparation includes:

• grinding the repair area and surrounding fiberglass
• filling fasteners and seams with body filler such as Bondo
• smoothing transitions between the existing surface and the repair area

These preparation steps ensure that the new fiberglass layer bonds properly with the existing deck surface.

Step 6 — Fiberglass Reinforcement Installation

After preparation is complete, fiberglass reinforcement is installed over the repaired section.

The fiberglass patch is applied using the same reinforcement system used for full deck installations.

This typically involves:

• fiberglass mat reinforcement
• resin saturation
• full coverage of the repaired area

The new fiberglass layer overlaps the surrounding fiberglass surface to ensure a continuous waterproof membrane.

Step 7 — Grinding and Surface Preparation

Once the fiberglass repair patch has cured, the repaired surface is ground to smooth any irregularities and prepare the area for finishing.

Grinding ensures that the repair blends seamlessly with the surrounding deck surface.

Proper surface preparation helps prevent visible transitions between the repaired area and the existing deck.

Step 8 — Surface Cleaning

After grinding is complete, the repaired area is cleaned thoroughly.

Dust and debris are removed using:

- blowing compressed air
- vacuuming the deck surface
- wiping the repair area with acetone

Cleaning ensures that the gelcoat finish will bond properly to the fiberglass surface.

Step 9 — Gelcoat Application

The final step in the soft spot repair process is the application of gelcoat.

The gelcoat restores the protective surface of the deck and seals the fiberglass repair patch.

Textured gelcoat finishes are typically applied to maintain slip resistance and match the surrounding deck surface.

Once the gelcoat cures, the repaired section becomes fully integrated into the waterproof fiberglass deck system.

Long-Term Repair Performance

When performed correctly, soft spot repairs restore the structural integrity and waterproof performance of the fiberglass deck.

By removing all deteriorated materials and installing new fiberglass reinforcement, the repaired area becomes fully protected against future water infiltration.

Routine inspection and maintenance help ensure that repaired areas continue to perform reliably over time.

Chapter 4

Structural Column Repair Systems

Structural columns are commonly used in residential decks to support upper-level decks, roof structures, or balcony systems. In many homes, these structural posts are concealed inside decorative column wraps constructed from PVC, Azek, or composite materials.

Because the structural post is hidden inside the column assembly, flashing failures around these posts may remain undetected for long periods of time.

When fiberglass flashing around the structural post is missing or improperly installed, water may infiltrate the deck system and cause deterioration of the surrounding plywood substrate.

In advanced cases, prolonged moisture exposure may also affect the structural framing beneath the deck.

Inspection and Access

When soft spots appear around a column base, the repair process begins by gaining access to the structural post.

In most cases, technicians begin by removing the **lower trim moldings** at the base of the column.

This allows inspection of the flashing condition around the structural post and evaluation of the surrounding plywood.

In many decks, railing systems are attached directly to the column assemblies. As a result, the railing system may need to be removed before the column trim can be disassembled.

If the damage extends beyond the flashing and affects the structural post itself, the entire decorative column assembly may need to be removed to allow full access to the structural post.

Structural Inspection

Once the column base is exposed, technicians evaluate several components of the deck structure, including:

• the plywood substrate surrounding the post
• fiberglass flashing around the post
• structural joists beneath the deck
• the structural post itself

The extent of repairs depends on the severity of water infiltration.

If the structural framing beneath the deck has been affected by prolonged moisture exposure, additional structural repairs may be required.

Column Flashing Repair

If the structural post remains in good condition and the problem was caused by missing or improperly installed flashing, the repair typically involves restoring the waterproof transition around the post.

This process generally includes:

• removing damaged plywood around the column base
• installing replacement plywood as required
• installing new fiberglass flashing around the structural post
• integrating the flashing with the deck waterproofing system

Once the fiberglass flashing has been installed and cured, the deck surface is restored using fiberglass reinforcement and gelcoat finishing.

After the waterproofing system has been restored, the decorative column components are reassembled.

Structural Post Replacement

If the structural post itself has been compromised by moisture damage, the repair becomes more extensive.

In these situations, the entire column assembly may need to be removed in order to replace the damaged structural post.

Once the damaged post has been replaced, the deck system is rebuilt with proper flashing installation to prevent future water infiltration.

Chapter 5

Railing Post Structural Repair Systems

Railing posts represent one of the most common penetrations in fiberglass deck systems.

Improper installation of railing posts can compromise the waterproof membrane and allow water to infiltrate the deck structure.

Many decks use railing systems that include decorative vinyl sleeves covering the structural post. Because the structural components are hidden inside these sleeves, water infiltration may not be immediately visible.

When water enters through improperly sealed post penetrations, it can cause deterioration of the plywood substrate surrounding the post location.

Removal of Damaged Posts

When a railing post becomes unstable or water infiltration has damaged the surrounding deck structure, the repair process begins by removing the existing post.

The fiberglass surface and plywood surrounding the post location are then cut to expose the underlying framing.

This allows technicians to inspect the condition of the deck structure and determine the extent of the repair.

Structural Reinforcement

Proper installation of railing posts requires adequate structural support beneath the deck surface.

During repairs, technicians typically install structural blocking between joists to provide a solid anchoring point for the new post.

Blocking is installed using lumber that matches the width of the deck joists.

This reinforcement helps distribute loads and improves the long-term stability of the railing system.

Installing Replacement Posts

After installing the necessary structural blocking, a new railing post is installed.

In many repair scenarios, **4x4 pressure-treated posts** are preferred because they provide strong structural support and reliable long-term performance.

The new post is securely fastened to the blocking and joists before the deck surface is rebuilt.

Rebuilding the Deck Surface

Once the post has been installed and secured, the plywood substrate is replaced and secured using appropriate fastening methods.

After the structural repairs are complete, the fiberglass waterproof membrane is restored around the post penetration.

Fiberglass flashing is installed around the base of the post to ensure that the penetration remains fully waterproof.

Finally, the repaired section of the deck is finished with gelcoat to restore the walking surface.

Chapter 6

Drip Edge Reconstruction Systems

The drip edge plays a critical role in directing water away from the deck structure.

In many older fiberglass decks, wooden drip edges were used instead of modern PVC materials. Over time, these wooden components may deteriorate due to prolonged moisture exposure.

Another common construction problem occurs when the original deck was built **without installing packer boards** beneath the plywood at the deck perimeter.

When packer boards are missing, exterior fascia boards may be installed flush with the drip edge.

Because the fascia and drip edge share the same thickness, the fascia may not fit beneath the drip edge as intended.

This condition prevents proper water drainage and allows water to infiltrate between the drip edge and the fascia board.

Consequences of Improper Drip Edge Installation

When water enters behind the fascia, moisture may accumulate around the structural framing at the deck perimeter.

Over time, this infiltration can lead to:

• deterioration of the plywood substrate
• moisture damage to structural joists
• hidden rot behind fascia boards

Because these structural components are concealed behind exterior trim, the damage may remain undetected until significant deterioration has occurred.

Drip Edge Reconstruction Procedure

When improper drip edge construction is identified, the repair process typically involves reconstructing the deck perimeter.

The standard repair procedure may include:

1. Removing the existing wooden drip edge.

2. Installing a **packer board** equal to the thickness of the removed drip edge.

3. Installing a larger **PVC drip edge**, commonly a **1x6 PVC board**.

The larger PVC drip edge extends beyond the fascia and ensures that water drains properly over the fascia surface.

This configuration restores the correct drainage path and prevents water from infiltrating behind the fascia.

Structural Repairs at the Deck Edge

If water infiltration has already damaged the structural joists behind the fascia, additional repairs may be required.

These repairs may include:

• removing fascia boards
• inspecting the structural joists
• replacing damaged sections of framing
• rebuilding the deck perimeter structure

Once the structural framing has been repaired, the new drip edge system is installed to restore proper water drainage.

Long-Term Edge Protection

Modern drip edge systems using PVC materials provide superior durability compared to older wood-based systems.

When properly installed with packer boards and correct drainage design, the deck perimeter can effectively direct water away from the structure and protect the deck framing from moisture damage.

Routine inspection of the deck edge helps identify potential drainage problems before structural deterioration occurs.

SEASHORE FIBERGLASS LLC

MASTER TECHNICAL MANUAL

Volume 4 — Operations, Estimating & Field Procedures

Chapter 1

Professional Deck Evaluation for Estimates

Accurate evaluation of fiberglass decks is essential before preparing any repair or restoration estimate. A proper inspection allows contractors to determine the true condition of the deck system and identify the underlying causes of deterioration.

Because water infiltration can travel beneath the fiberglass membrane, the visible damage on the deck surface may not represent the original source of the problem. For this reason, professional deck evaluation requires a systematic inspection process.

The purpose of the evaluation is to determine:

- the structural condition of the deck
- the condition of the fiberglass membrane
- the condition of waterproof transitions
- the extent of required repairs

A correct diagnosis ensures that the recommended repair solution addresses the root cause of the problem rather than only the visible symptoms.

Initial Client Consultation

The evaluation process typically begins with a brief discussion with the homeowner or property owner.

Clients may provide valuable information regarding the history of the deck, including:

- when the deck was originally installed
- previous repairs performed on the deck
- areas where leaks have been observed
- any structural concerns noticed by the homeowner

Although this information can provide helpful context, the technician must rely primarily on physical inspection of the deck system.

Visual Inspection Procedure

The first stage of the field evaluation involves a visual inspection of the entire deck surface.

Technicians observe the deck carefully to identify visible signs of deterioration, including:

• large cracks in the fiberglass membrane
• clusters of small cracks (spider cracking)
• worn or faded gelcoat surfaces
• surface discoloration caused by moisture
• damaged or deteriorated drip edges

These visual indicators may reveal areas where water infiltration has already begun.

However, visual inspection alone is not sufficient to determine the full condition of the deck system.

Physical Inspection

After completing the visual inspection, technicians perform a physical inspection by walking across the deck surface.

Walking the deck allows the technician to identify structural weakness beneath the fiberglass membrane.

Areas that feel soft, unstable, or spongy may indicate deterioration of the plywood substrate.

Technicians apply pressure while walking across the deck to detect soft spots that may not be immediately visible.

Special attention is given to areas near:

• walls connected to the deck
• door transitions
• railing posts
• decorative columns
• deck perimeter edges

These locations are common points where water infiltration occurs.

Inspection of Critical Waterproofing Transitions

In addition to evaluating the deck surface, technicians must carefully inspect all transitions where the fiberglass membrane meets vertical structures.

Critical transition points include:

• wall connections
• door thresholds
• railing posts
• structural columns
• deck perimeter edges

These transitions rely on proper flashing installation to maintain waterproof integrity.

Improper flashing installation is one of the most common causes of fiberglass deck leaks.

Determining the Appropriate Repair Solution

Once the inspection has been completed, the technician must determine the most appropriate repair or restoration method.

Fiberglass decks generally fall into one of three service categories:

1. Recolor maintenance
2. Reglass restoration
3. Structural reconstruction

The correct recommendation depends on the condition of both the fiberglass membrane and the underlying deck structure.

Recolor Maintenance Evaluation

Recolor maintenance is appropriate when the fiberglass membrane remains structurally sound but the gelcoat surface has become worn.

Typical indicators that recolor maintenance may be sufficient include:

• faded or worn gelcoat finish
• minor surface wear without fiberglass damage
• limited cosmetic cracking

Recolor maintenance restores the protective gelcoat surface and extends the service life of the fiberglass system.

Reglass Restoration Evaluation

Reglass restoration is recommended when the fiberglass membrane has deteriorated significantly but the deck structure remains stable.

Indicators that reglass may be required include:

- widespread cracking across the deck surface
- aged fiberglass reinforcement
- numerous previous repair patches
- deteriorated drip edge components

Reglass restoration installs a new fiberglass reinforcement layer across the entire deck surface.

This process renews the waterproof membrane while preserving the existing structural deck system.

Structural Reconstruction Evaluation

In severe cases, water infiltration may cause extensive deterioration of the plywood substrate and structural framing.

Indicators that structural reconstruction may be necessary include:

- multiple large soft spots across the deck
- severely deteriorated plywood
- structural joists affected by rot
- noticeable instability in the deck structure

When structural damage reaches this level, surface restoration methods such as recolor or reglass may not be sufficient.

Instead, portions of the deck structure must be rebuilt before installing a new fiberglass waterproof system.

Importance of Accurate Estimation

Proper deck evaluation allows contractors to prepare accurate repair estimates and recommend the most appropriate solution for the client.

Misdiagnosing the condition of a fiberglass deck can lead to incomplete repairs and recurring problems.

By following a systematic inspection process, technicians can identify the true causes of deck deterioration and implement effective repair strategies.

Accurate evaluation not only protects the contractor's reputation but also ensures long-term performance of the fiberglass deck system.

Chapter 2

Determining the Correct Service: Recolor, Reglass, or Structural Reconstruction

One of the most important responsibilities during deck evaluation is determining the appropriate level of repair required for the fiberglass deck system.

Choosing the correct service ensures that the underlying problems are properly addressed while avoiding unnecessary work.

Fiberglass deck services generally fall into three primary categories:

- Recolor Maintenance
- Reglass Restoration
- Structural Reconstruction

Each of these services addresses different levels of deck deterioration.

Understanding the differences between them allows technicians to recommend the correct solution for each project.

Recolor Maintenance

Recolor maintenance is the least invasive service and is typically used when the fiberglass membrane remains structurally sound.

Over time, exposure to sunlight, foot traffic, and environmental conditions causes the gelcoat surface to wear down. When this occurs, the fiberglass beneath the gelcoat may still remain intact.

In these cases, recolor maintenance restores the protective surface layer without replacing the fiberglass reinforcement.

Indicators that recolor maintenance is appropriate include:

- faded or worn gelcoat finish
- minor cosmetic cracking
- small surface imperfections without structural damage
- absence of soft spots in the deck surface

Recolor maintenance restores the deck's protective coating and helps extend the lifespan of the fiberglass system.

Reglass Restoration

Reglass restoration is required when the fiberglass membrane itself has begun to deteriorate.

This condition often occurs after many years of exposure to environmental conditions or when the deck has undergone multiple previous repairs.

During reglass restoration, a new fiberglass reinforcement layer is installed across the entire deck surface.

Indicators that reglass restoration may be required include:

- widespread cracking across the deck
- aging fiberglass reinforcement
- multiple previous repair patches
- exposed fiberglass fibers
- deteriorated gelcoat combined with fiberglass fatigue

Reglass restoration renews the waterproof membrane and provides long-term protection for the deck system.

Structural Reconstruction

Structural reconstruction becomes necessary when water infiltration has significantly damaged the deck structure.

In these situations, the plywood substrate and sometimes the structural framing beneath the deck have deteriorated.

Indicators that structural reconstruction may be required include:

- multiple soft spots across the deck surface
- severely deteriorated plywood
- structural joists affected by moisture damage
- noticeable sagging or instability in the deck structure

When structural damage reaches this stage, surface restoration methods alone are not sufficient.

Structural components must be repaired or replaced before a new fiberglass waterproof system can be installed.

Importance of Correct Diagnosis

Selecting the appropriate repair method requires careful inspection and professional judgment.

Applying recolor maintenance to a deck that requires reglass restoration may result in premature failure.

Similarly, performing reglass restoration without addressing underlying structural damage may lead to recurring problems.

Proper diagnosis ensures that the repair solution addresses the root cause of the deck's deterioration.

Chapter 3

Field Work Planning and Job Preparation

Successful fiberglass deck projects require careful planning before any work begins.

Proper preparation allows technicians to complete projects efficiently while maintaining high installation standards.

Planning begins once the repair scope has been determined during the inspection and estimating process.

Job Site Preparation

Before work begins, the job site must be prepared to allow safe and efficient access to the deck area.

Preparation tasks may include:

- clearing furniture and obstacles from the deck surface
- protecting surrounding areas from dust and debris
- establishing safe access points for technicians and materials

Preparing the work environment reduces delays and helps prevent damage to surrounding property.

Removal of Existing Components

Many fiberglass deck projects require temporary removal of deck components before repairs can begin.

Common components removed during preparation include:

- railing systems
- decorative trim elements
- fascia boards
- siding sections when flashing repairs are required

Removing these components allows technicians to access the full deck surface and critical waterproofing transitions.

Once the deck repairs are completed, these components can be reinstalled.

Material Preparation

Fiberglass deck work requires specialized materials and tools.

Before beginning the installation process, technicians ensure that all required materials are available on site.

Typical materials include:

- fiberglass reinforcement mat
- resin systems

- body filler materials
- replacement plywood panels
- PVC drip edge components
- gelcoat finishing materials

Proper material preparation helps ensure that work progresses smoothly without unnecessary interruptions.

Weather Considerations

Weather conditions can significantly affect fiberglass deck work.

Temperature, humidity, and precipitation can influence the curing process of fiberglass resin and gelcoat materials.

Technicians must monitor weather conditions carefully when scheduling fiberglass installation.

Work may need to be delayed during periods of heavy rain or extremely low temperatures to ensure proper curing of materials.

Chapter 4

Safety Procedures for Fiberglass Deck Work

Working with fiberglass materials requires careful attention to safety procedures.

Fiberglass installation and repair involve tools, chemicals, and elevated work environments that can present potential hazards if proper precautions are not followed.

Maintaining safe working conditions protects both technicians and property owners.

Personal Protective Equipment

Technicians working with fiberglass materials should use appropriate protective equipment.

Recommended protective equipment includes:

• safety glasses
• gloves resistant to resin exposure
• respiratory protection when grinding fiberglass
• protective clothing to prevent skin irritation

These safety measures help protect technicians from fiberglass particles, dust, and chemical exposure.

Safe Use of Power Tools

Fiberglass deck repair work frequently involves the use of power tools such as grinders, saws, and drills.

Technicians must operate these tools carefully and follow proper safety procedures.

Equipment should be inspected regularly to ensure it remains in safe working condition.

Working at Elevated Heights

Many fiberglass decks are located on elevated structures such as second-story decks or balconies.

When working at height, technicians must ensure that proper fall protection measures are in place.

Maintaining stable working platforms and safe access points reduces the risk of accidents.

Chemical Handling

Fiberglass resin and gelcoat materials contain chemicals that must be handled properly.

Technicians should follow manufacturer safety guidelines when mixing and applying these materials.

Adequate ventilation should be maintained whenever possible to minimize exposure to fumes.

Chapter 5

Client Communication and Professional Recommendations

Clear communication with clients is an important part of professional fiberglass deck services.

Homeowners may not fully understand the technical aspects of deck deterioration or the causes of leaks.

Technicians must explain the condition of the deck and the recommended repair solution in a clear and professional manner.

Explaining Deck Problems to Clients

When discussing deck issues with clients, technicians should explain the underlying causes of deterioration.

Examples may include:

- improper flashing installation
- aging fiberglass membranes
- water infiltration through deck penetrations
- deterioration of drip edge systems

Providing clear explanations helps clients understand why certain repairs are necessary.

Presenting Repair Options

In some situations, multiple repair options may be available.

For example, a deck may qualify for either reglass restoration or full structural reconstruction depending on the extent of deterioration.

Technicians should present the available options along with the advantages and limitations of each solution.

This allows the client to make an informed decision regarding the repair approach.

Setting Realistic Expectations

Clients should also understand the expected performance and maintenance requirements of fiberglass deck systems.

Technicians may explain that fiberglass decks typically require:

• recolor maintenance approximately every five years
• reglass restoration approximately every ten years depending on conditions

Providing realistic expectations helps maintain long-term client satisfaction.

Professional Reputation

High-quality work and honest communication help build trust between contractors and clients.

By providing accurate evaluations and reliable repair solutions, contractors establish a reputation for professionalism and technical expertise.

This reputation is essential for long-term success in fiberglass deck services.

SEASHORE FIBERGLASS LLC

MASTER TECHNICAL MANUAL

Volume 5 — Knowledge Base & Expert FAQ

Chapter 1

Understanding Fiberglass Waterproof Deck Systems

Fiberglass decks are one of the most durable waterproof deck systems when installed correctly. Unlike traditional wood decks that rely on surface coatings or membranes applied over finished structures, fiberglass decks function as a fully integrated waterproof system.

The fiberglass membrane forms a continuous, seamless barrier that prevents water from penetrating into the deck structure.

This system combines several layers that work together to create a long-lasting waterproof deck surface.

These layers typically include:

- structural framing
- plywood substrate layers
- fiberglass reinforcement membrane
- gelcoat protective surface

Each layer plays a specific role in maintaining the durability and waterproof performance of the deck system.

Structural Deck Framing

The structural framing of the deck supports the entire deck system and carries the loads generated by occupants, furniture, and environmental conditions.

Deck framing is typically constructed using pressure-treated lumber and may include:

- joists
- beams
- posts
- ledger connections to the house structure

Proper framing design ensures that the deck remains stable and maintains the correct slope required for water drainage.

A standard slope used in fiberglass decks is approximately:

1/4 inch per foot

This slope allows water to drain efficiently toward the deck perimeter or drainage system.

Plywood Substrate System

The plywood substrate provides the base surface onto which the fiberglass membrane is installed.

Many fiberglass decks use a two-layer plywood system that improves structural rigidity and reduces movement in the deck surface.

A common configuration includes:

• first layer: 3/4 inch CDX plywood
• second layer: 1/2 inch ACX plywood

The first layer may be installed using construction adhesive and nails, while the second layer is typically glued and screwed for improved structural stability.

Reducing movement in the plywood substrate is essential for preventing cracks in the fiberglass membrane.

Fiberglass Reinforcement Layer

The fiberglass reinforcement layer forms the primary waterproof barrier of the deck system.

Fiberglass mat is installed across the deck surface and saturated with resin to create a rigid composite layer.

Once cured, this fiberglass layer becomes a continuous waterproof membrane.

A common reinforcement system includes:

2-ounce fiberglass mat

This material provides a balance between flexibility and structural strength.

The fiberglass membrane must be installed seamlessly across the deck surface to ensure full waterproof protection.

Gelcoat Protective Layer

The gelcoat layer forms the outer surface of the fiberglass deck.

Gelcoat provides several important functions, including:

• protection against ultraviolet radiation
• resistance to moisture penetration

- durable walking surface
- aesthetic appearance

Gelcoat finishes often include textured surfaces to improve slip resistance, particularly in outdoor environments where the deck may become wet.

Periodic maintenance of the gelcoat layer helps extend the life of the fiberglass deck system.

Chapter 2

Why Fiberglass Decks Fail

Although fiberglass decks are highly durable, failures can occur when installation errors, aging materials, or lack of maintenance affect the system.

Most fiberglass deck failures are not caused by the fiberglass material itself but rather by problems at waterproof transitions or penetrations.

Understanding these failure mechanisms helps technicians diagnose problems more effectively.

Improper Flashing Installation

One of the most common causes of fiberglass deck leaks is improper flashing installation.

Flashings are required wherever the deck surface meets vertical structures such as:

- walls
- door thresholds
- posts
- columns

If fiberglass flashing is missing or improperly installed, water can enter beneath the fiberglass membrane and spread throughout the deck system.

Because flashing areas are critical transition points, careful installation is required to maintain waterproof integrity.

Post Penetrations

Railing posts and structural posts represent penetrations through the fiberglass membrane.

If these penetrations are not properly sealed and flashed, water may infiltrate the deck structure through fastener holes or gaps around the post base.

Movement in railing posts may also cause cracking in the fiberglass surrounding the post penetration.

Proper installation of fiberglass flashing around posts is essential for preventing leaks.

Drip Edge Failures

The deck perimeter must be designed to allow water to drain away from the deck structure.

Improper drip edge construction can allow water to infiltrate behind fascia boards and into structural framing.

This problem often occurs when packer boards are not installed beneath the deck perimeter.

Without packer boards, fascia boards may sit flush with the drip edge, preventing proper water drainage.

Over time, water infiltration behind fascia boards can lead to hidden structural damage.

Aging Fiberglass Membranes

Fiberglass membranes are durable but eventually degrade after long-term exposure to environmental conditions.

As the fiberglass system ages, the gelcoat surface may wear down and the fiberglass reinforcement may develop cracks.

When cracking becomes widespread across the deck surface, the membrane may require reglass restoration.

Routine maintenance can help extend the service life of fiberglass deck systems.

Chapter 3

Common Misconceptions About Fiberglass Decks

Many homeowners have misconceptions about fiberglass deck systems.

Clarifying these misunderstandings helps clients make better decisions about maintenance and repairs.

Misconception: Fiberglass Decks Never Require Maintenance

Some homeowners believe fiberglass decks require no maintenance.

In reality, the gelcoat surface gradually wears over time due to exposure to sunlight, weather, and foot traffic.

Periodic recolor maintenance helps restore the protective surface and extend the lifespan of the fiberglass system.

Misconception: All Deck Cracks Require Full Replacement

Not all cracks indicate that the deck must be completely rebuilt.

Small surface cracks may be repaired through localized fiberglass repairs or reglass restoration depending on the overall condition of the deck.

Professional inspection is required to determine the appropriate repair method.

Misconception: Water Always Appears Where the Leak Starts

Many homeowners assume that the visible leak location represents the source of the problem.

In fiberglass decks, water may travel beneath the membrane following the slope of the deck.

As a result, the location where damage appears may be far from the original point of water entry.

Proper inspection is required to identify the true leak source.

Chapter 4

Recommended Maintenance for Fiberglass Deck Systems

Routine maintenance helps extend the lifespan of fiberglass deck systems and prevents minor issues from developing into major repairs.

Recommended maintenance practices include:

- regular cleaning of the deck surface
- periodic inspection of flashing transitions
- monitoring railing post stability
- inspecting the deck perimeter for drainage problems

Professional inspection every few years can help detect early warning signs of deterioration.

Maintenance Intervals

Typical maintenance intervals may include:

Recolor maintenance: approximately every 5 years

Reglass restoration: approximately every 10 years depending on environmental exposure and deck condition.

These intervals may vary depending on climate conditions and deck usage.

Chapter 5

Professional Fiberglass Deck Standards

High-quality fiberglass deck systems require proper installation techniques and careful attention to detail.

Professional standards include:

• correct deck slope for drainage
• stable plywood substrate installation
• proper flashing installation at all transitions
• correct drip edge construction
• seamless fiberglass membrane installation

When these standards are followed, fiberglass decks can provide long-term waterproof performance.

SEASHORE FIBERGLASS LLC

MASTER TECHNICAL MANUAL

Volume 6 — Expert Questions & Answers

Chapter 1

General Questions About Fiberglass Decks

What is a fiberglass deck?

A fiberglass deck is a waterproof deck system that uses fiberglass reinforcement and resin to create a seamless waterproof membrane over a structural deck surface.

The fiberglass membrane is installed over a plywood substrate and then sealed with a protective gelcoat finish.

Once cured, the fiberglass layer forms a rigid waterproof barrier that prevents water from penetrating into the deck structure.

This system is commonly used for balconies, rooftop decks, and elevated exterior decks where waterproofing is essential.

Why are fiberglass decks commonly used in coastal areas?

Fiberglass decks are particularly well suited for coastal environments because they provide excellent resistance to moisture, salt air, and weather exposure.

In regions such as the Jersey Shore, decks are exposed to:

- high humidity
- salt air
- rain and snow
- temperature fluctuations

Traditional wood decks often deteriorate quickly under these conditions.

Fiberglass decks provide a durable waterproof surface that protects the structural deck system from water infiltration.

How long does a fiberglass deck last?

A properly installed fiberglass deck can last for many years when maintained correctly.

Typical maintenance cycles include:

Recolor maintenance: approximately every 5 years

Reglass restoration: approximately every 10 years depending on environmental exposure.

The actual lifespan of the deck depends on several factors, including installation quality, environmental conditions, and maintenance practices.

Regular maintenance helps extend the lifespan of the fiberglass system.

Are fiberglass decks completely waterproof?

Yes. When installed correctly, fiberglass decks are fully waterproof.

The fiberglass membrane forms a continuous surface that prevents water from penetrating into the deck structure.

However, waterproof performance depends heavily on proper installation of flashing and transition details.

If flashings around walls, doors, or posts are improperly installed, water may still infiltrate the system.

Why do fiberglass decks crack?

Fiberglass decks may develop cracks for several reasons.

The most common causes include:

- movement in the plywood substrate
- aging fiberglass reinforcement
- improper installation of plywood layers
- structural movement in the deck framing

Small surface cracks may be cosmetic, but larger cracks can allow water infiltration if not repaired.

Professional inspection is required to determine the severity of cracking.

Chapter 2

Questions About Deck Leaks

Why is my fiberglass deck leaking?

Leaks in fiberglass decks are usually caused by failures at waterproof transitions rather than failure of the fiberglass surface itself.

Common leak sources include:

- wall flashings
- door flashings
- railing posts
- structural columns
- drip edge failures

Water may enter through these areas and travel beneath the fiberglass membrane before visible damage appears.

Why do I have a soft spot in my deck?

Soft spots occur when the plywood substrate beneath the fiberglass membrane becomes weakened due to prolonged moisture exposure.

When water infiltrates the deck system and remains trapped beneath the fiberglass layer, the plywood may begin to deteriorate.

Soft spots are typically detected when walking across the deck and feeling areas that are soft or unstable underfoot.

Repair usually requires removing the damaged plywood and restoring the fiberglass membrane.

Why does the damage appear far from where the leak started?

Water inside a fiberglass deck system often travels beneath the membrane following the slope of the deck.

Most decks are constructed with a slope of approximately 1/4 inch per foot.

When water enters the deck system at a higher point, such as near a wall or door, it may travel downward across the deck before causing visible damage.

This is why the visible damage may appear far from the original leak source.

Chapter 3

Questions About Deck Repairs

What is a reglass restoration?

Reglass restoration is a repair method that installs a new fiberglass reinforcement layer across the entire deck surface.

This process is typically performed when the fiberglass membrane has aged and developed widespread cracking.

Reglass restoration renews the waterproof membrane without requiring complete reconstruction of the deck structure.

What is the difference between recolor and reglass?

Recolor maintenance restores the gelcoat surface of the deck without replacing the fiberglass reinforcement layer.

Reglass restoration installs a new fiberglass reinforcement layer over the deck.

Recolor is used when the fiberglass membrane remains structurally sound.

Reglass is used when the fiberglass reinforcement has deteriorated and requires renewal.

Can soft spots be repaired without replacing the entire deck?

Yes. Many soft spots can be repaired through localized repairs.

This typically involves:

- removing the damaged fiberglass and plywood
- replacing the affected plywood
- installing new fiberglass reinforcement
- restoring the gelcoat surface

However, if multiple soft spots are present across the deck, larger restoration work may be required.

Chapter 4

Questions About Maintenance

How often should fiberglass decks be maintained?

Fiberglass decks should be inspected regularly and maintained as needed.

Typical maintenance intervals include:

Recolor maintenance every 5 years.

Reglass restoration approximately every 10 years depending on the deck condition.

Routine inspection helps detect early warning signs before major repairs are required.

How can I extend the life of my fiberglass deck?

Several maintenance practices can help extend the lifespan of fiberglass decks.

Recommended practices include:

• keeping the deck surface clean
• inspecting flashing transitions periodically
• repairing cracks early
• maintaining the gelcoat surface

Professional inspection every few years can help identify potential problems before they become serious.

Chapter 5

Questions About Deck Construction

Why is proper deck slope important for fiberglass decks?

Proper deck slope is essential because it allows water to drain efficiently off the deck surface.

Most fiberglass decks are built with a slope of approximately **1/4 inch per foot**. This pitch ensures that water does not accumulate on the deck surface.

If a deck is built without sufficient slope, water may pond on the surface, which can eventually lead to premature deterioration of the gelcoat and fiberglass membrane.

Correct slope design is one of the most important structural factors in long-term fiberglass deck performance.

Why are two layers of plywood often used in fiberglass decks?

Many fiberglass decks use two layers of plywood because this configuration increases structural rigidity and reduces movement in the deck surface.

Movement in the plywood substrate can cause cracking in the fiberglass membrane.

A typical plywood system may include:

• first layer: 3/4 inch CDX plywood
• second layer: 1/2 inch ACX plywood

The second layer is usually installed using adhesive and screws to stabilize the surface.

A stable substrate helps prevent fiberglass cracking and extends the lifespan of the waterproof system.

Why are screws preferred instead of nails for the top plywood layer?

Screws provide a stronger and more stable connection between the plywood and the deck framing.

Older decks were often constructed using nails, which may loosen over time.

When nails loosen, the plywood may move slightly, which can cause cracks in the fiberglass membrane.

Using screws helps prevent this movement and provides a more secure deck surface.

What is the purpose of chamfer strips where the deck meets the wall?

Chamfer strips create a smooth transition between the horizontal deck surface and vertical wall surfaces.

Without chamfer strips, the fiberglass would need to bend sharply at a 90-degree angle.

Sharp angles can weaken fiberglass reinforcement and increase the risk of cracking.

Chamfer strips create a gradual transition that allows the fiberglass to bond properly and maintain structural strength.

Chapter 6

Questions About Deck Edges and Drainage

What is a drip edge?

A drip edge is a structural component installed along the perimeter of the deck that directs water away from the deck structure.

The drip edge allows water to flow off the deck and prevents water from running back toward the deck framing.

Modern fiberglass deck systems often use **PVC or Azek drip edges**, which resist moisture and do not rot.

Why did older decks often use wooden drip edges?

Older fiberglass decks were sometimes constructed using wooden drip edges because PVC materials were not as widely available at the time.

However, wood is vulnerable to moisture exposure.

Over time, wooden drip edges may deteriorate and allow water to infiltrate the deck perimeter.

This deterioration can eventually affect the plywood substrate and structural framing.

Why is a packer board installed at the deck perimeter?

A packer board creates the correct spacing between the plywood deck surface and the exterior fascia.

The packer board is typically narrower than the drip edge, which allows the exterior fascia to fit underneath the drip edge.

This design ensures that water flows **over the fascia instead of behind it**.

Without the packer board, fascia boards may sit flush with the drip edge, which can allow water to infiltrate behind the fascia.

What happens if the packer board was not installed?

If the packer board was not installed during deck construction, the fascia may sit flush with the drip edge.

This condition prevents proper drainage and allows water to infiltrate between the fascia and the drip edge.

Over time, this moisture may reach the structural framing behind the fascia.

In severe cases, structural joists may begin to deteriorate.

Repairing this condition often requires reconstructing the deck edge using a new packer board and larger PVC drip edge.

Chapter 7

Questions About Deck Flashings

What is flashing in a fiberglass deck system?

Flashing is a waterproof transition installed where the fiberglass membrane meets vertical structures.

Flashings prevent water from entering the deck system at critical transition points.

These transitions include:

- wall connections
- door thresholds
- railing posts
- structural columns

Proper flashing installation is essential for maintaining a waterproof deck system.

Why are flashing failures so common?

Flashing failures often occur because fiberglass decks require specialized installation knowledge.

Many contractors who are unfamiliar with fiberglass systems may incorrectly install flashing details.

If flashing is missing or improperly integrated with siding or doors, water can infiltrate beneath the fiberglass membrane.

Because flashing areas are often hidden behind siding or trim, problems may remain undetected for long periods.

Why do leaks often start near doors?

Doors represent a critical transition point between the building envelope and the deck surface.

If flashing beneath the door threshold is not properly installed, water can infiltrate beneath the deck system.

Because doors are often located at the higher elevation of the deck, water entering at this point may travel downward across the deck before causing visible damage.

Why must siding sometimes be removed to repair flashing?

In many cases, siding covers the flashing transition between the wall and the deck.

If flashing is missing or improperly installed, the siding must be temporarily removed to access the wall structure.

Removing the siding allows technicians to install proper flashing that integrates with the house wrap and fiberglass membrane.

Once the flashing repair is complete, the siding can be reinstalled.

Chapter 8

Questions About Railing Posts and Columns

Why are railing posts common sources of leaks?

Railing posts penetrate the fiberglass membrane.

If these penetrations are not properly sealed and flashed, water can enter through the fastener holes or gaps around the post.

Over time, this infiltration can cause deterioration of the plywood surrounding the post location.

Movement in the railing system can also cause cracks in the fiberglass around the post base.

What are structural mount posts?

Structural mount posts are railing posts that are mounted on top of the deck surface using mechanical fasteners.

These posts are typically attached using screws or bolts that penetrate the deck surface.

If the penetrations are not properly sealed with fiberglass flashing, water may infiltrate the deck system through these fasteners.

Proper flashing around the post base is necessary to maintain waterproofing.

Why are 4x4 pressure-treated posts often more stable?

4x4 pressure-treated posts are anchored directly to the deck framing.

This installation method provides strong structural support and reduces movement in the railing system.

Because these posts are connected directly to the structural framing, they often provide greater long-term stability compared to surface-mounted posts.

However, proper waterproof flashing must still be installed around the post penetration.

Why are leaks around columns difficult to detect?

Structural posts inside decorative columns are often hidden behind PVC or composite column wraps.

Because the structural post is concealed, flashing problems around the post base may not be visible.

Water infiltration can continue for long periods before visible damage appears.

By the time soft spots develop around the column base, the damage may already be extensive.

Chapter 9

Questions About Structural Deck Damage

How do structural joists become damaged in fiberglass decks?

Structural joists are typically damaged when water infiltration continues for long periods without repair.

Water entering through flashing failures or drip edge problems may eventually reach the structural framing beneath the deck.

Once moisture reaches the joists, wood rot may begin to develop.

Over time, this deterioration can weaken the structural capacity of the deck.

What are the warning signs of structural damage?

Common warning signs include:

- multiple soft spots across the deck
- visible sagging in the deck surface
- instability when walking on the deck
- deterioration visible from beneath the deck

When these symptoms appear, structural inspection is required.

When does a deck require full reconstruction?

Full reconstruction may be required when:

- the plywood substrate is severely deteriorated
- structural joists have rotted
- the deck structure has become unstable

In these situations, repairing only the fiberglass surface will not solve the underlying structural problem.

The deck structure must be repaired before installing a new waterproof system.

Chapter 10

Questions About Professional Fiberglass Deck Services

Why should fiberglass decks be repaired by specialists?

Fiberglass decks require specialized installation techniques and knowledge.

Many contractors are experienced with traditional deck construction but may not be familiar with fiberglass waterproofing systems.

Improper installation of fiberglass components can lead to leaks and structural damage.

Experienced fiberglass deck contractors understand how to properly install flashing systems, waterproof penetrations, and structural deck components.

Why is professional inspection important?

Professional inspection allows technicians to identify hidden problems that may not be visible to homeowners.

Because water can travel beneath the fiberglass membrane, visible damage does not always indicate the true source of the problem.

A thorough inspection helps ensure that repairs address the root cause of the issue.

What makes a fiberglass deck system durable?

Several factors contribute to the durability of fiberglass decks.

These include:

- proper structural framing
- stable plywood substrate installation
- correct flashing details
- proper fiberglass reinforcement installation
- regular maintenance of the gelcoat surface

When these factors are properly managed, fiberglass decks can provide reliable waterproof protection for many years.

Chapter 11

Advanced Technical Questions About Fiberglass Deck Systems

Why is grinding the deck surface essential before applying fiberglass?

Grinding the deck surface is one of the most critical preparation steps in fiberglass installation.

Fiberglass resin requires a rough surface to achieve proper mechanical adhesion. If fiberglass is applied over a smooth or contaminated surface, the bond between the new fiberglass layer and the existing substrate may fail.

Grinding removes:

- old contaminants
- loose fiberglass fibers
- worn gelcoat surfaces
- dust and debris

This process creates a properly prepared bonding surface that allows the new fiberglass layer to adhere securely.

Why must the fiberglass surface also be ground after the fiberglass layer cures?

Once fiberglass reinforcement cures, the surface may contain small irregularities, texture variations, or excess resin.

Grinding the cured fiberglass surface ensures that the gelcoat finish will adhere properly.

Without grinding, the gelcoat may not bond correctly to the fiberglass surface, which can lead to premature peeling or surface failure.

This step also helps smooth the surface and prepare it for the final finish.

Why is acetone used before applying gelcoat?

Acetone is commonly used to clean the fiberglass surface before applying gelcoat.

Grinding operations produce dust, oils, and contaminants that may remain on the surface.

Acetone removes these contaminants and ensures that the gelcoat bonds properly to the fiberglass layer.

Proper cleaning is essential to ensure long-term adhesion of the finish layer.

Why are fiberglass decks typically finished with textured gelcoat?

Textured gelcoat improves traction on the deck surface.

Because fiberglass decks are exposed to rain and moisture, smooth surfaces may become slippery.

A textured finish provides:

• improved safety when the deck is wet
• enhanced durability of the gelcoat surface
• a visually appealing finish

Textured gelcoat is the standard finish for most fiberglass deck systems.

Why is fiberglass mat commonly used instead of fiberglass cloth?

Fiberglass mat provides multidirectional reinforcement, which helps distribute loads evenly across the deck surface.

Unlike woven fiberglass cloth, fiberglass mat contains randomly oriented fibers that create a strong composite structure when saturated with resin.

This reinforcement method allows the fiberglass membrane to resist cracking caused by structural movement or environmental stress.

For deck applications, fiberglass mat is typically preferred.

Chapter 12

Common Contractor Mistakes in Fiberglass Deck Installation

What is one of the most common installation mistakes?

One of the most common mistakes is improper flashing installation.

Many contractors focus on installing the fiberglass membrane but fail to properly integrate flashing systems around walls, doors, and structural penetrations.

Without proper flashing, water can enter beneath the fiberglass membrane even if the fiberglass itself is installed correctly.

This mistake often leads to leaks and long-term structural damage.

Why is improper deck slope a serious problem?

Deck slope is essential for proper drainage.

If a deck is built without sufficient slope, water may accumulate on the surface.

Standing water can accelerate deterioration of the gelcoat and fiberglass membrane.

Over time, this condition may lead to cracks, water infiltration, and structural damage.

Correct deck slope must be established during the framing stage.

Why is installing fiberglass over unstable plywood a major mistake?

Fiberglass membranes rely on a stable substrate.

If the plywood beneath the fiberglass moves due to poor fastening or structural instability, cracks may form in the fiberglass layer.

Many older decks were installed using nails rather than screws.

If the plywood becomes loose, it must be re-secured before installing new fiberglass reinforcement.

Ignoring substrate movement can lead to premature system failure.

Why is improper installation of railing posts a common problem?

Railing posts often penetrate the fiberglass membrane.

If contractors install posts using screws or bolts without proper flashing and sealing, water may infiltrate the deck system through these penetrations.

Over time, this infiltration may cause plywood deterioration and soft spots around the post location.

Proper flashing around post penetrations is essential for maintaining waterproof integrity.

Why is skipping surface preparation a major mistake?

Some inexperienced contractors attempt to apply fiberglass or gelcoat without properly preparing the surface.

Skipping grinding, cleaning, or surface repairs may cause poor adhesion between layers.

This can lead to peeling gelcoat, delamination of fiberglass, and premature failure of the deck system.

Proper preparation is essential for long-term performance.

Chapter 13

Fiberglass Deck Challenges in Coastal Environments

Why are fiberglass decks commonly used in coastal regions?

Coastal environments present challenging conditions for outdoor structures.

These conditions include:

- high humidity
- salt air exposure
- frequent rainfall
- strong sunlight
- seasonal temperature changes

Fiberglass decks provide a durable waterproof system that can resist these environmental conditions better than many traditional deck materials.

How does salt air affect deck systems?

Salt air contains microscopic salt particles that can accelerate corrosion and deterioration of many materials.

Metal fasteners, railings, and structural components may be affected by salt exposure.

Fiberglass systems are highly resistant to salt air, which makes them well suited for coastal installations.

However, metal components used in deck construction should be corrosion-resistant whenever possible.

Why do coastal decks require more maintenance?

Coastal decks are exposed to harsh environmental conditions that can accelerate wear.

Salt air, ultraviolet radiation, and moisture can gradually degrade the gelcoat surface.

Regular maintenance helps preserve the protective finish and extend the lifespan of the fiberglass membrane.

Routine inspection is particularly important in coastal environments.

Why are drainage systems especially important near the ocean?

Coastal storms and heavy rainfall can place significant stress on deck drainage systems.

Proper deck slope and drip edge design ensure that water flows away from the deck structure.

If drainage systems fail, water infiltration may lead to structural damage.

Proper drainage design is essential for long-term performance in coastal regions.

Why must flashing details be especially precise in coastal construction?

Because coastal structures are frequently exposed to wind-driven rain, flashing systems must be installed carefully to prevent water infiltration.

Wind can force water into small gaps that may not cause problems in milder environments.

Proper integration between fiberglass flashing, siding, house wrap, and door systems is essential for preventing leaks.